

3. Sensor Layer

Internet of Things

Executive Summary

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Motivation: IoT systems require proper design perspectives and component selection to effectively interact with the physical world.

Problem: It is necessary to understand IoT design approaches, classify sensors and actuators, and relate them to real-world applications and hardware implementations.

Overview:

- Introduction to IoT design perspectives.
- Classification of sensors and actuators.
- Discussion of real-world IoT applications.
- Hands-on lab: Wheatstone bridge.

Conclusion: IoT design integrates system-level perspectives, device selection, and signal conditioning to enable efficient real-world applications.

Outline

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Introduction

Design perspective

Sensors and actuators

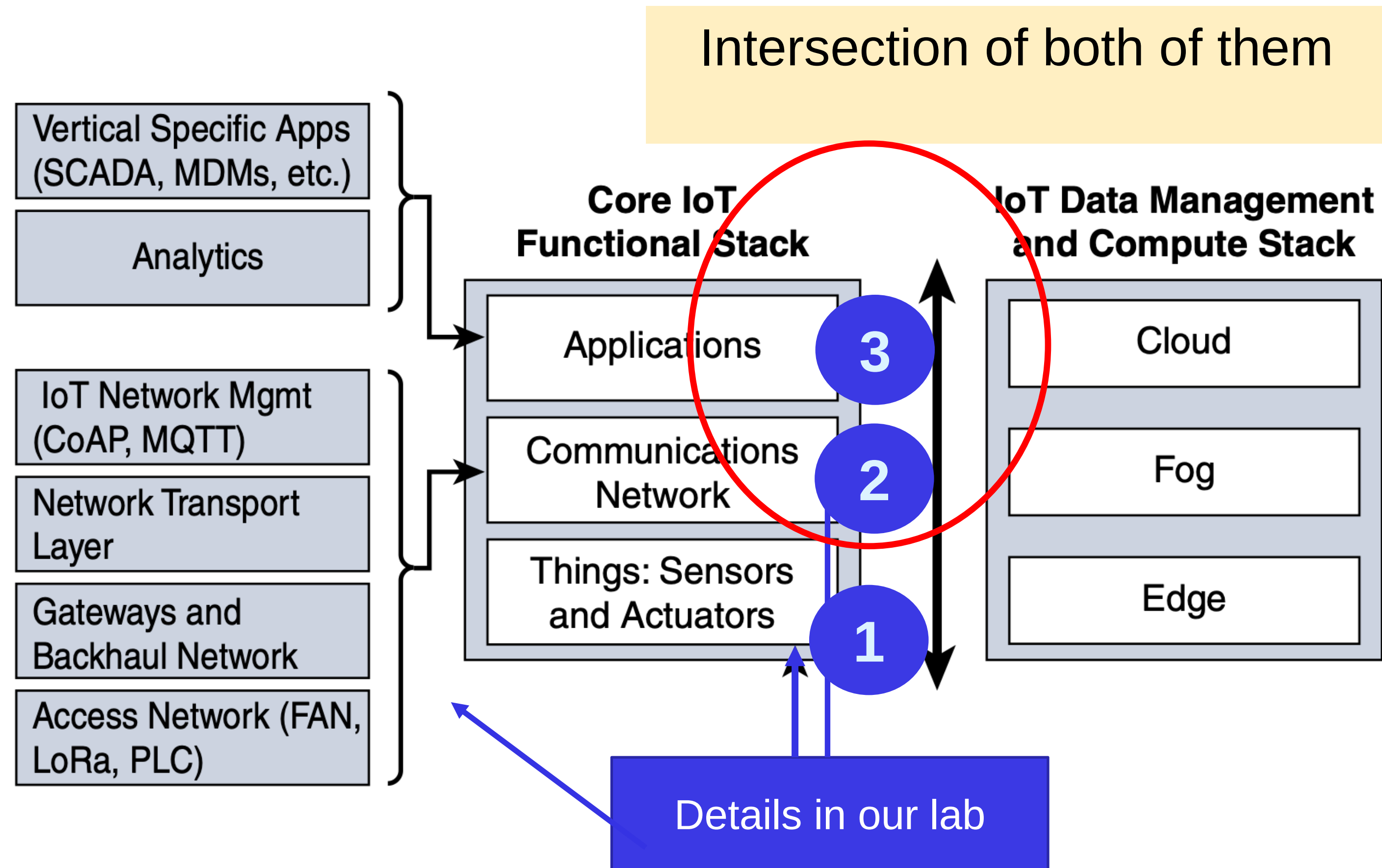
Case Study

Wheatstone bridge

Conclusions

Recall: General IoT Architecture

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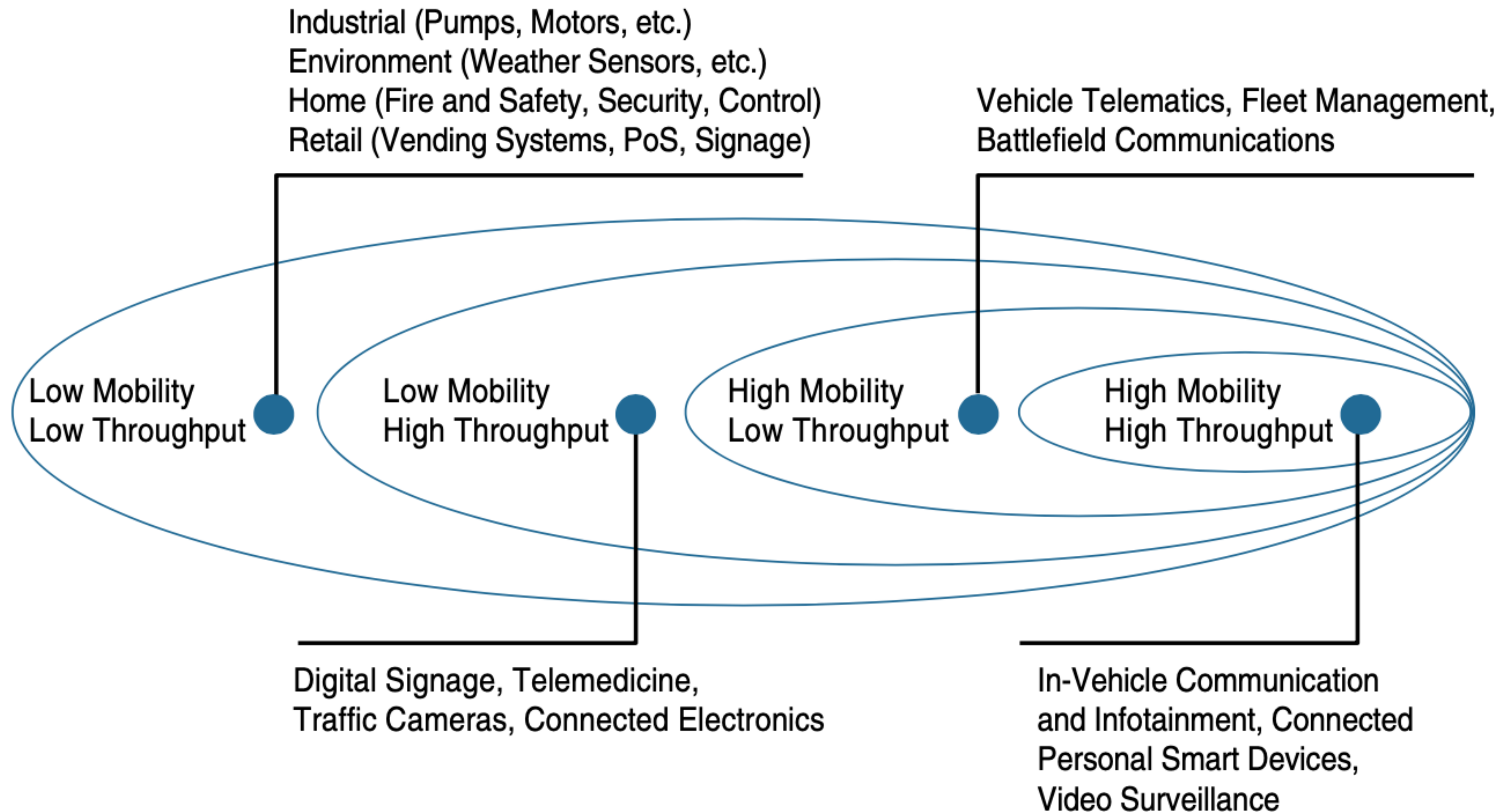


- **Layered architecture (abstraction to implementation)**
- **Current IoT target** has a very important component of analytics.
- Hardware sensing platforms **are not envisioned for computing intensive tasks.**
- **Offload computing:** defines the Compute Stack.

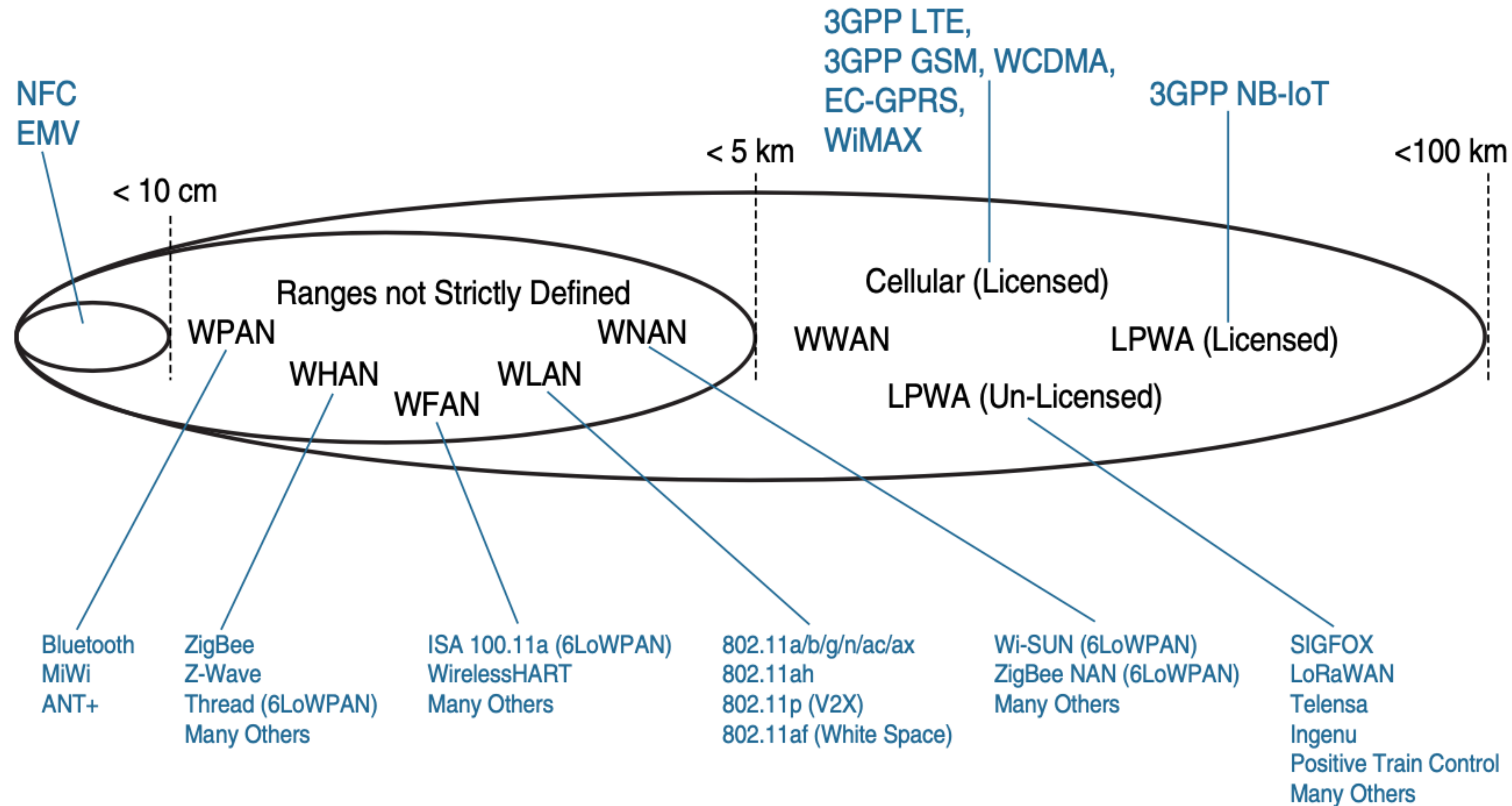
Sensor Applications

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- Mobility required by the sensor:



Distance: communication standards



WPAN: Wireless Personal Area Network

WHAN: Wireless Home Area Network

WFAN: Wireless Field (or Factory) Area Network

WLAN: Wireless Local Area Network

WNAN: Wireless Neighborhood Area Network

WWAN: Wireless Wide Area Network

LPWA: Low Power Wide Area

Outline

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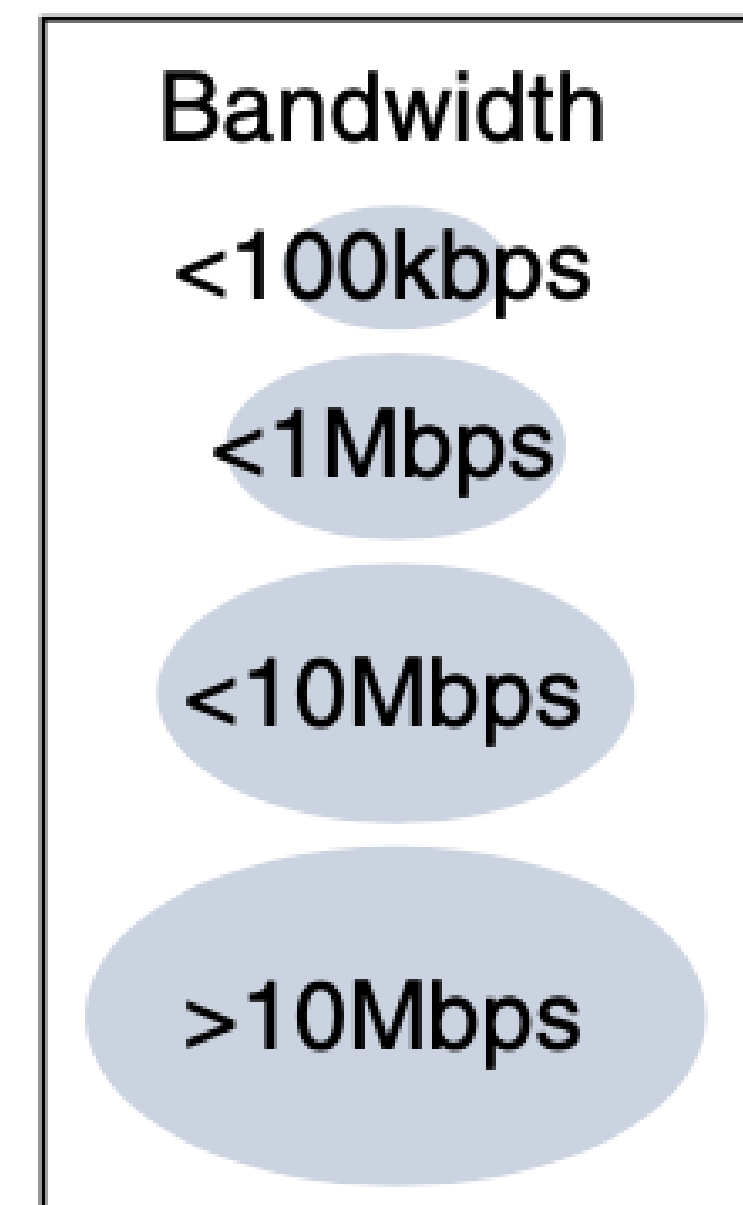
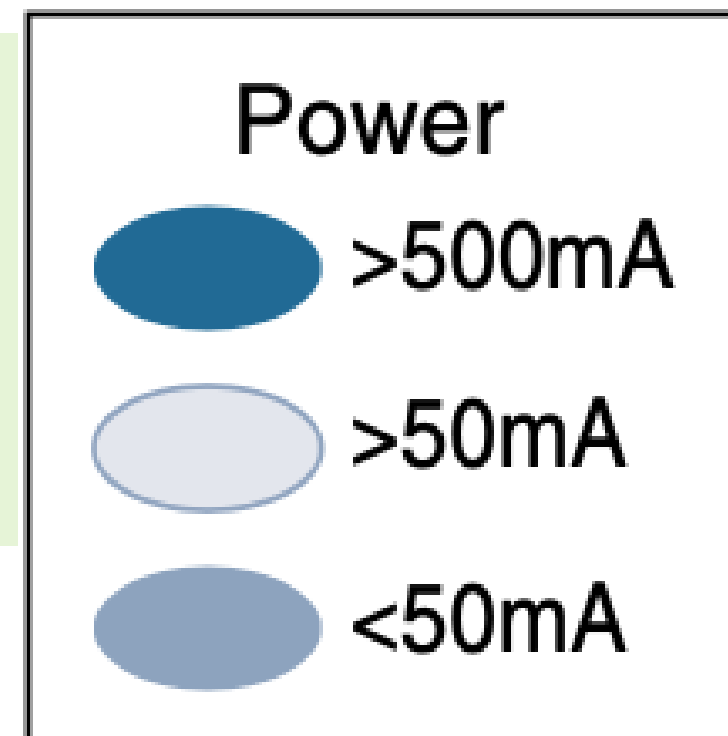
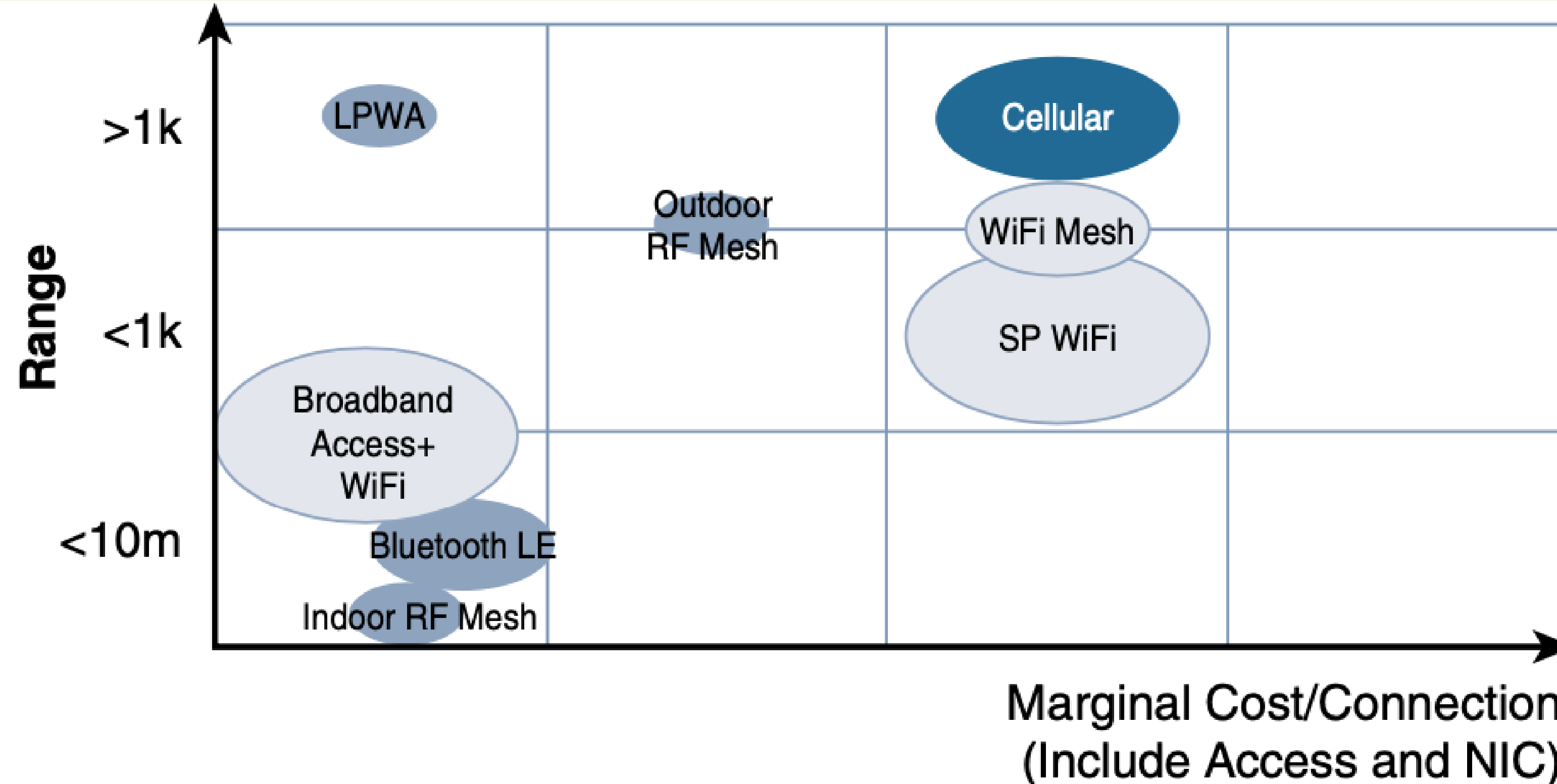
Case Study

Wheatstone bridge

Conclusions

Dinstance: Power and bandwidth

- **More is more?**
- **Desirable: High performance with low power operation**

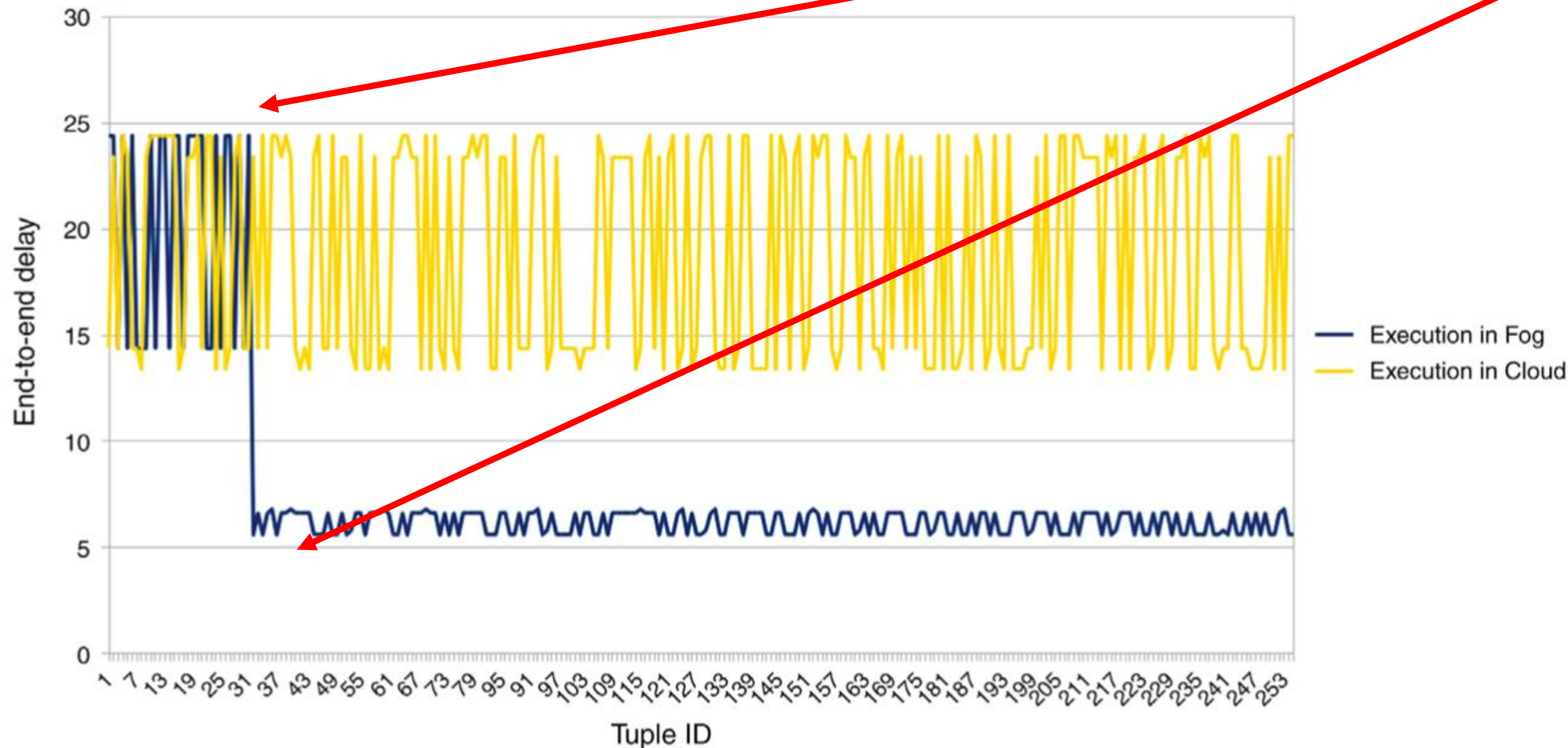


Performance in the computing stack

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Assume a tuple as pair of data from the IoT domain

Comparison of end-to-end tuple execution delays



What is desirable?
Fog vs Cloud
Higher delay or lower delay? Why?

Reasons to have a higher delay:

- Location (distance) for data availability
- Communication infrastructure
- Protocols (intermediate)

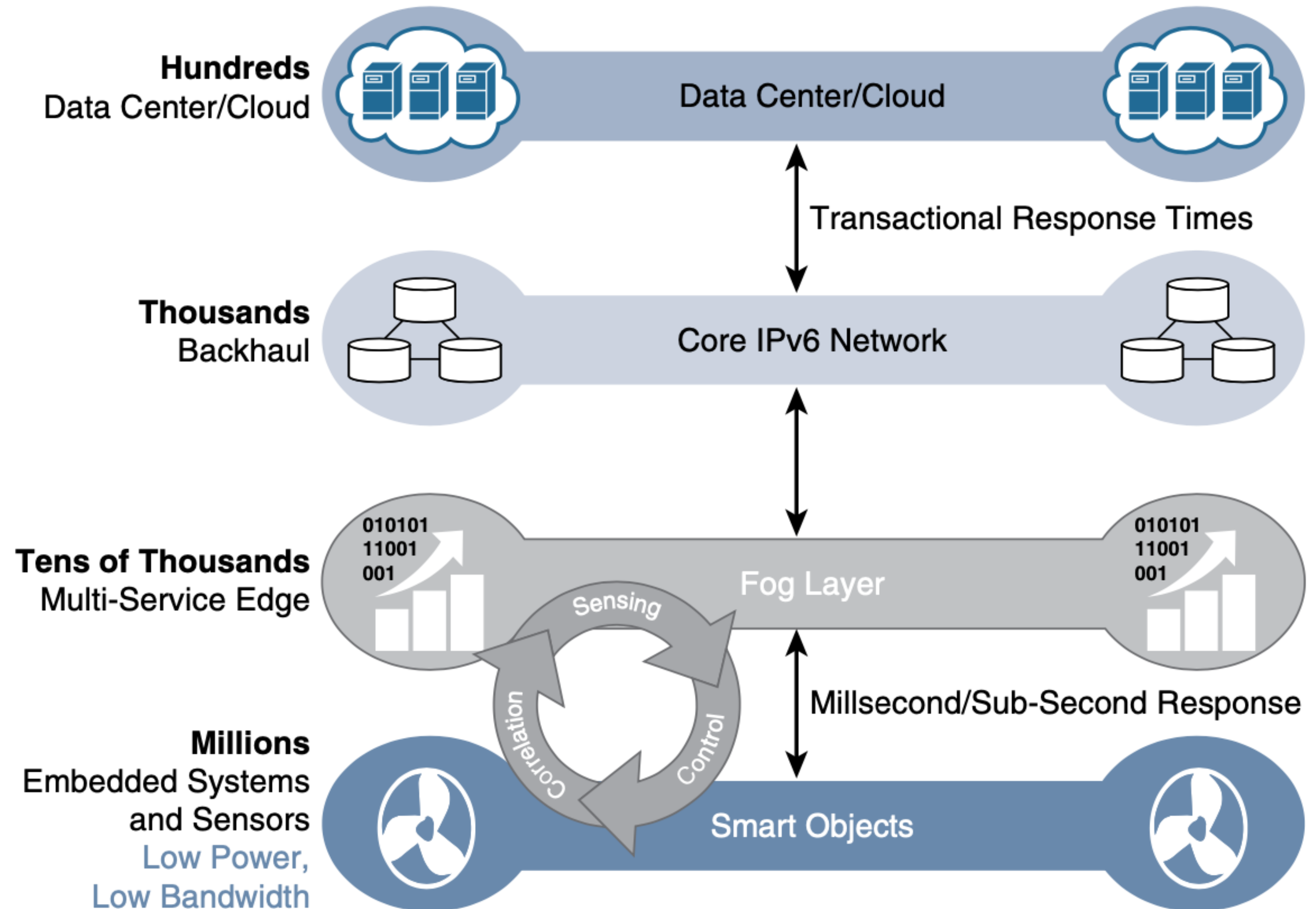
So... what is the limit of Fog?

Design perspective

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Design Principles:

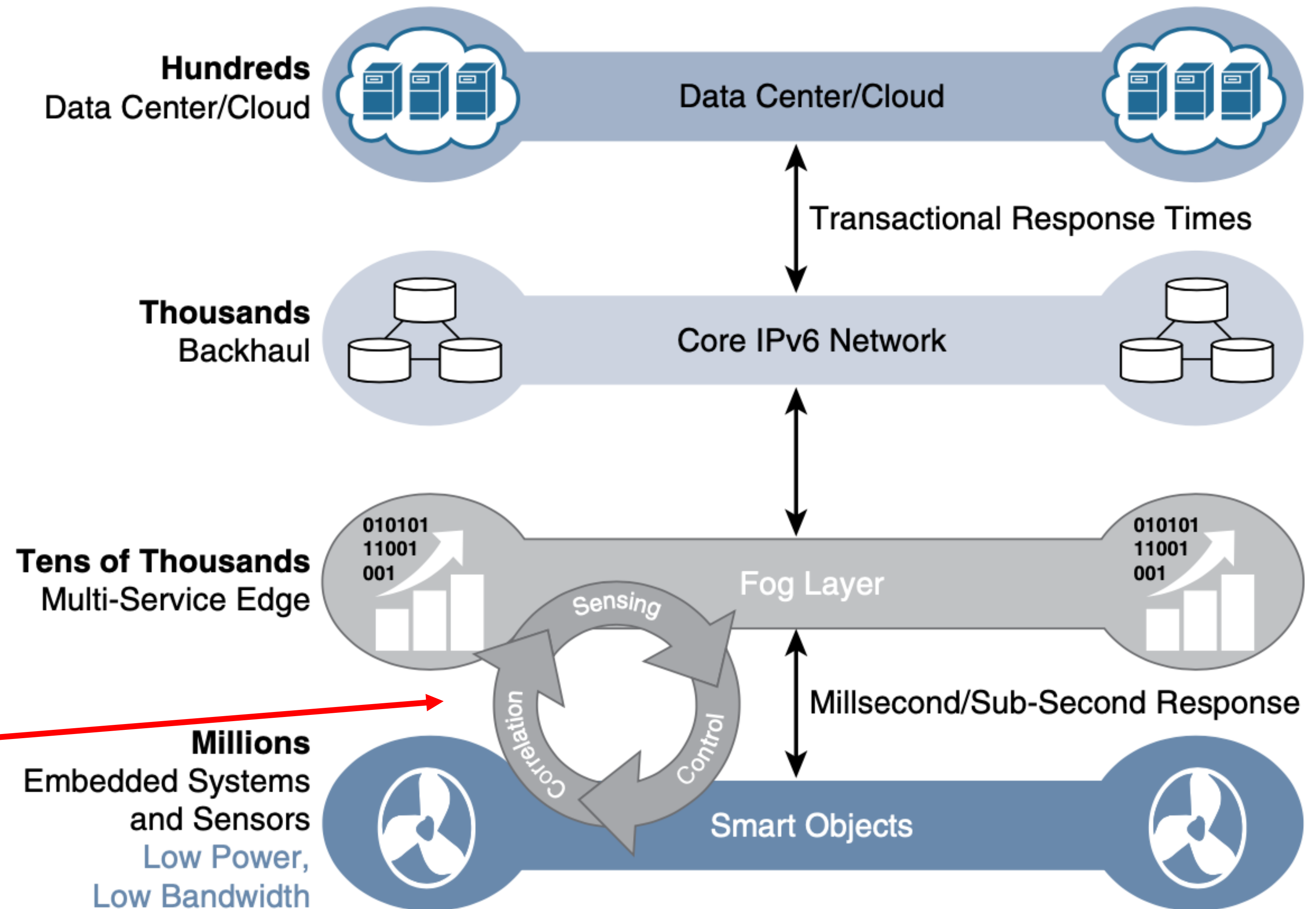
1. Minimizing latency
2. Conserving network bandwidth
3. Increasing local efficiency



Relevance of smart objects or things

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- **To reduce computing in the higher layer.**
- **Lower layer provides “smart” data.**
- **Things are required to be smart.**
- **Simplify data analysis, if possible: preprocessed data.**



Ready Data

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1. Only essential data to be processed in higher layers.
2. Smaller chunks of data in the network domain, reduce communication latency.

Idea:

- Maintain a lower layer of smart objects with limited networking and processing (functionality) characteristics.
- Lower layer data, does not require extreme security as higher layers.
- Security is one the main concerns for modern IoT services (open problem). Because the limited functionality is not focused on security.

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Transduction in IoT

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Bridging the Physical and Digital World

Transduction: The process of converting one form of energy (physical parameter) into another (electrical signal), or vice versa.

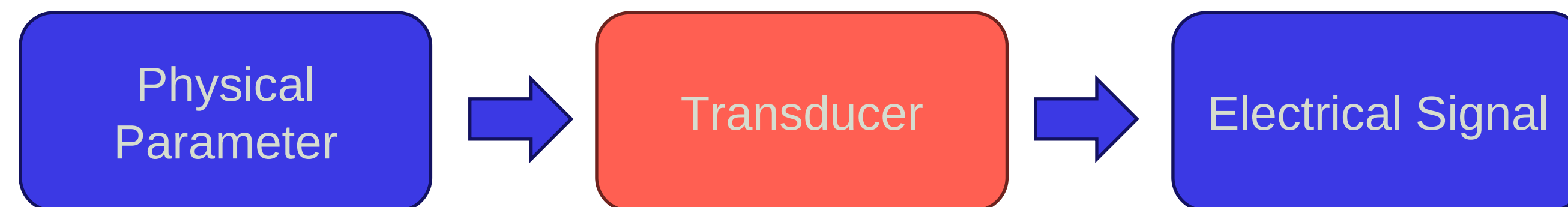
- **Physical World:** Temperature, pressure, light, acceleration.
- **Digital World:** Voltage, current, frequency, resistance.

Microcontrollers do not understand physical phenomena—they only read voltage and current changes. Transduction is the universal translator.

Sensors

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- **Definition:** Measures some physical quantity and converts that measurement reading into a digital representation.
- That digital representation is typically passed to another device for transformation into useful data that can be consumed by intelligent devices or humans.



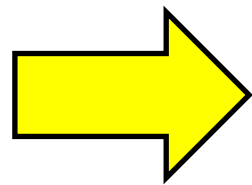
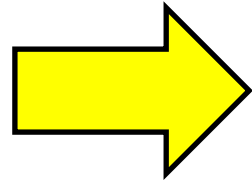
Sensor Classification

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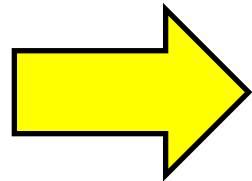
Active or passive	based on whether they produce an energy output and typically require an external power supply (active) or whether they simply receive energy and typically require no external power supply (passive).
Invasive or non-invasive	based on whether a sensor is part of the environment it is measuring (invasive) or external to it (non-invasive).
Contact or no-contact	based on whether they require physical contact with what they are measuring (contact) or not (no-contact).
Absolute or relative	based on whether they measure on an absolute scale (absolute) or based on a difference with a fixed or variable reference value (relative).
Area of application	based on the specific industry or vertical where they are being used.
How sensors measure	based on the physical mechanism used to measure sensory input (for example, thermoelectric, electrochemical, piezoresistive, optic, electric, fluid mechanic, photoelastic).
What sensors measure	based on their applications or what physical variables they measure.

Sensor Types

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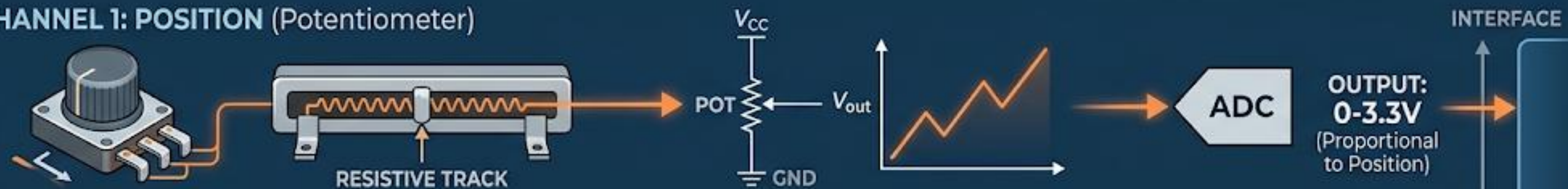
Sensor Types	Description	Examples
Position	A position sensor measures the position of an object; the position measurement can be either in absolute terms (absolute position sensor) or in relative terms (displacement sensor). Position sensors can be linear, angular, or multi-axis.	Potentiometer, inclinometer, proximity sensor
Occupancy and motion	Occupancy sensors detect the presence of people and animals in a surveillance area, while motion sensors detect movement of people and objects. The difference between the two is that occupancy sensors generate a signal even when a person is stationary, whereas motion sensors do not.	Electric eye, radar
Velocity and acceleration	Velocity (speed of motion) sensors may be linear or angular, indicating how fast an object moves along a straight line or how fast it rotates. Acceleration sensors measure changes in velocity.	Accelerometer, gyroscope
Force	Force sensors detect whether a physical force is applied and whether the magnitude of force is beyond a threshold.	Force gauge, viscometer, tactile sensor (touch sensor)
Pressure	Pressure sensors are related to force sensors, measuring force applied by liquids or gases. Pressure is measured in terms of force per unit area.	Barometer, Bourdon gauge, piezometer
Flow	Flow sensors detect the rate of fluid flow. They measure the volume (mass flow) or rate (flow velocity) of fluid that has passed through a system in a given period of time.	Anemometer, mass flow sensor, water meter



Sensor Types	Description	Examples
Acoustic	Acoustic sensors measure sound levels and convert that information into digital or analog data signals.	Microphone, geophone, hydrophone
Humidity	Humidity sensors detect humidity (amount of water vapor) in the air or a mass. Humidity levels can be measured in various ways: absolute humidity, relative humidity, mass ratio, and so on.	Hygrometer, humistor, soil moisture sensor
Light	Light sensors detect the presence of light (visible or invisible).	Infrared sensor, photodetector, flame detector
Radiation	Radiation sensors detect radiation in the environment. Radiation can be sensed by scintillating or ionization detection.	Geiger-Müller counter, scintillator, neutron detector
Temperature	Temperature sensors measure the amount of heat or cold that is present in a system. They can be broadly of two types: contact and non-contact. Contact temperature sensors need to be in physical contact with the object being sensed. Non-contact sensors do not need physical contact, as they measure temperature through convection and radiation.	Thermometer, calorimeter, temperature gauge
Chemical	Chemical sensors measure the concentration of chemicals in a system. When subjected to a mix of chemicals, chemical sensors are typically selective for a target type of chemical (for example, a CO ₂ sensor senses only carbon dioxide).	Breathalyzer, olfactometer, smoke detector
Biosensors	Biosensors detect various biological elements, such as organisms, tissues, cells, enzymes, antibodies, and nucleic acid.	Blood glucose biosensor, pulse oximetry, electrocardiograph

SENSOR TRANSDUCTION PRINCIPLES: MECHANISMS & INTERFACING

CHANNEL 1: POSITION (Potentiometer)



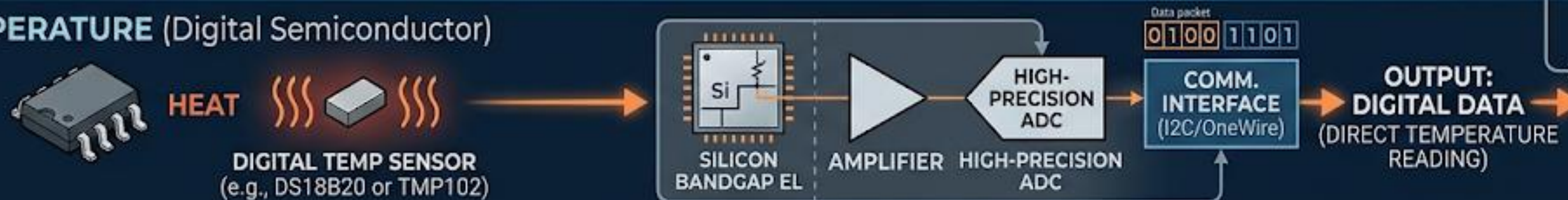
CHANNEL 2: VELOCITY (Hall Effect)



TEMPERATURE (Analog Resistive NTC)



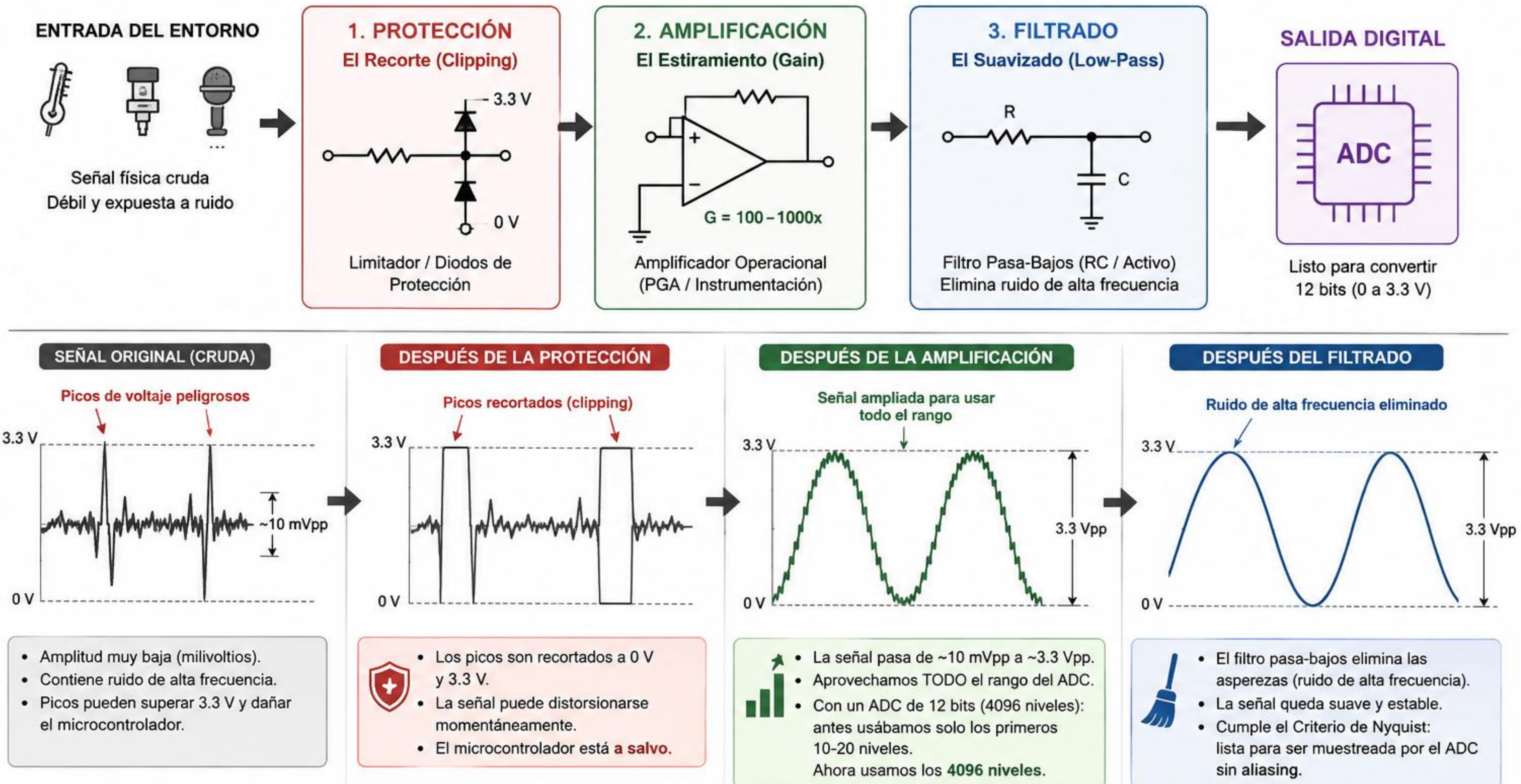
TEMPERATURE (Digital Semiconductor)



Signal Conditioning

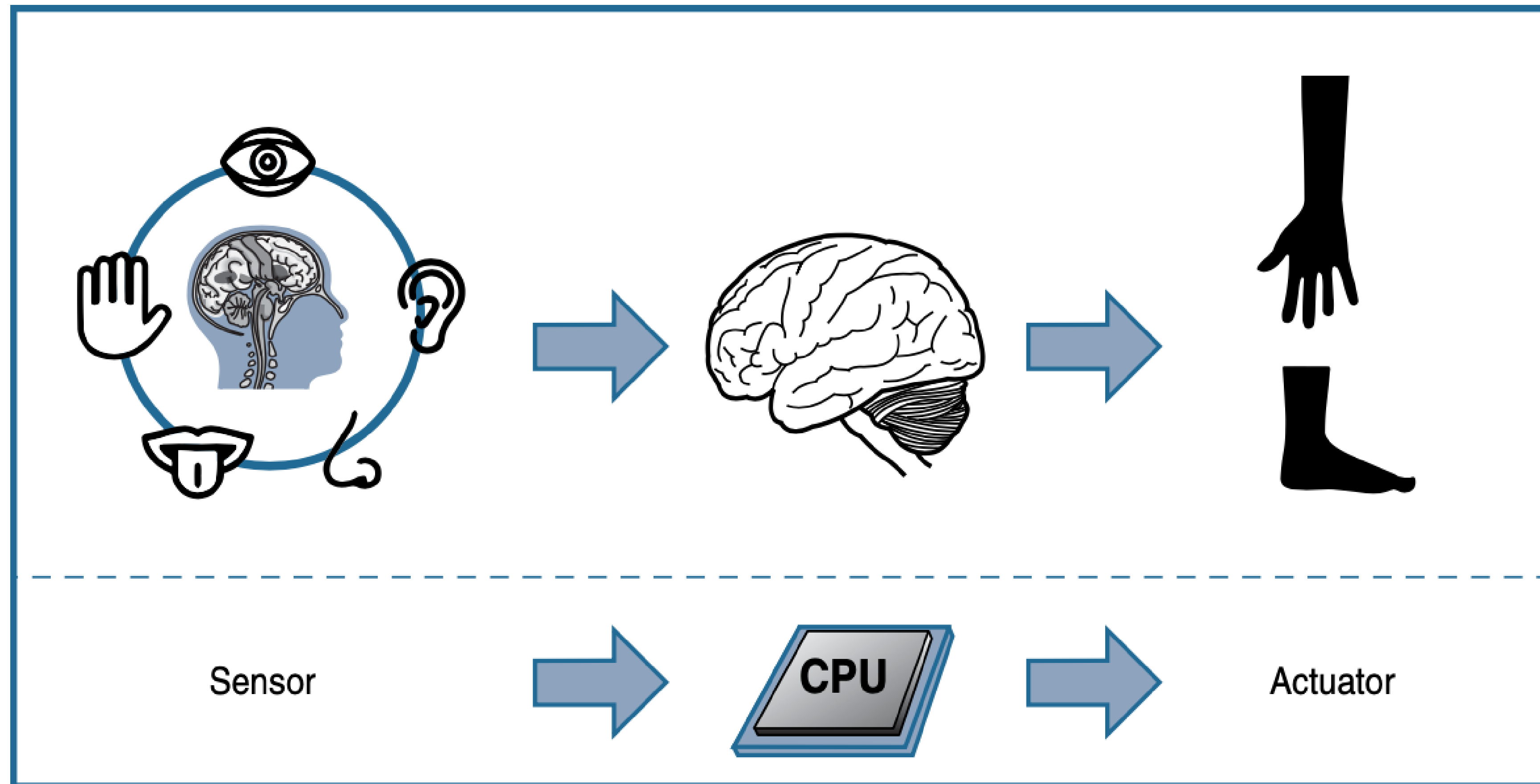
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- **Amplification:** Scales weak sensor voltages to match the microcontroller's dynamic range (0 - 3.3V) using Op-Amps.
- **Filtering:** Removes high-frequency noise and mains interference (50/60 Hz) using low-pass passive/active filters.
- **Protection & Isolation:** Prevents overvoltage and spikes using Zener diodes, voltage dividers, and optocouplers.



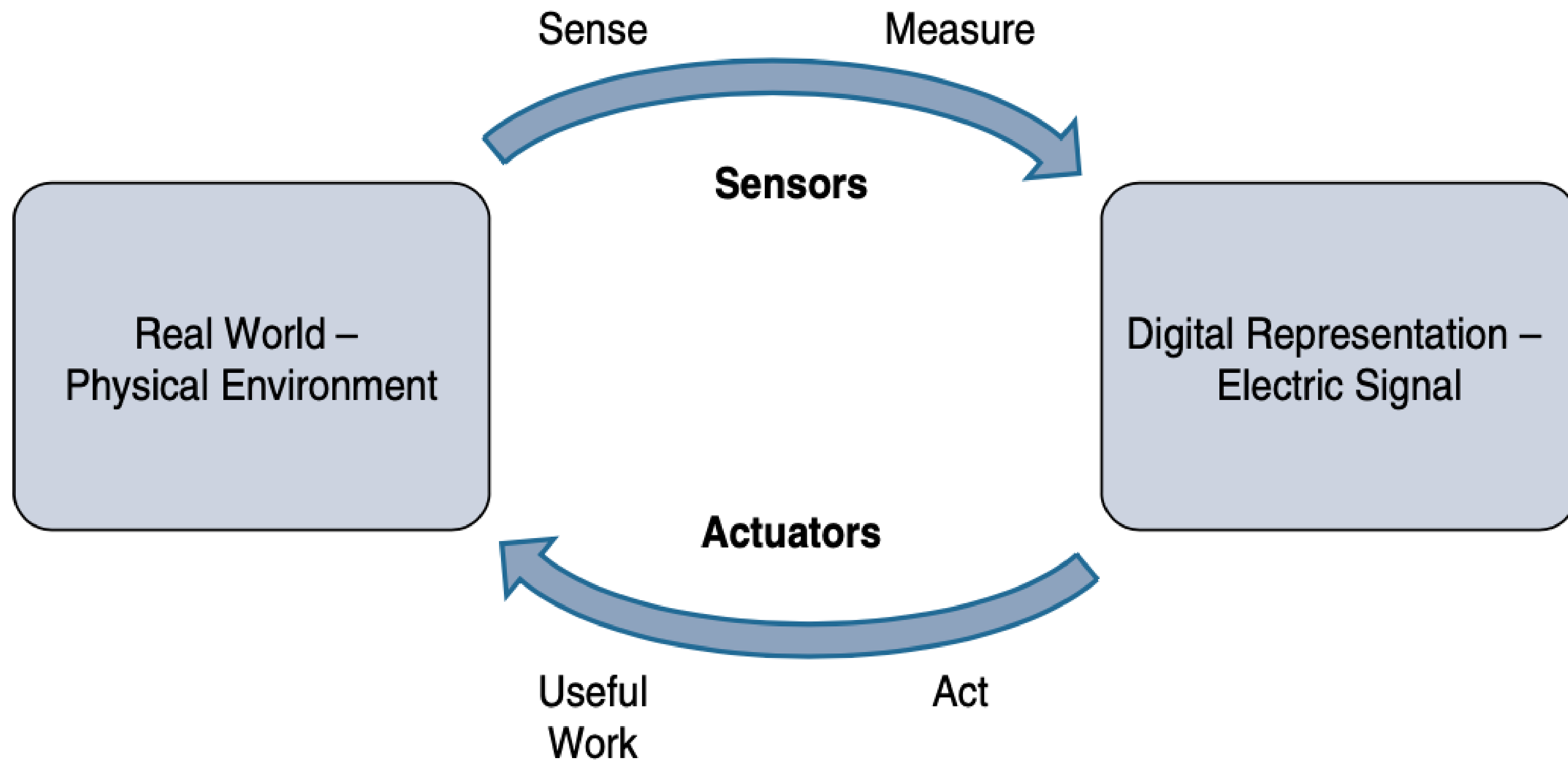
Human actuators

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Actuators

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Actuator classification

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- **Type of motion:** based on the type of motion they produce (for example, linear, rotary, one/two/three-axes).
- **Power:** based on their power output (for example, high power, low power, micro power)
- **Binary or continuous:** based on the number of stable-state outputs.
- **Area of application:** based on the specific industry or vertical where they are used.
- **Type of energy:** Actuators can be classified based on their energy type.

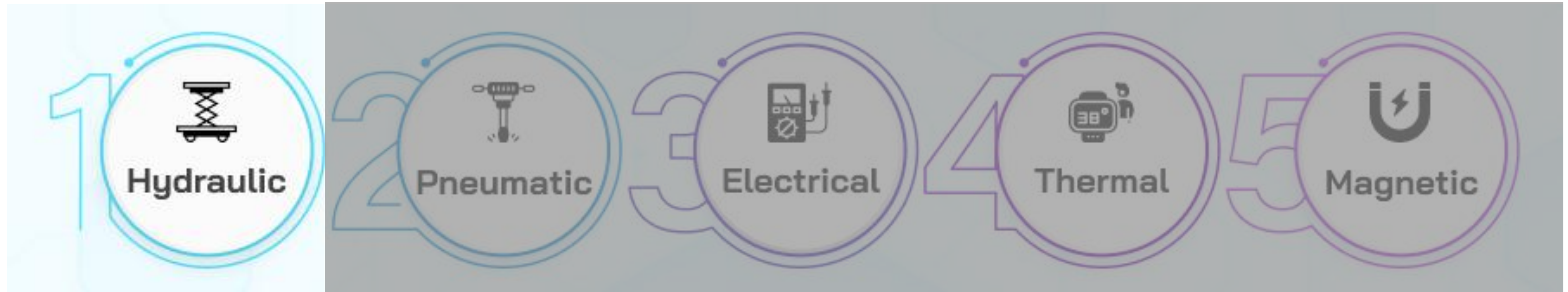
Actuator types

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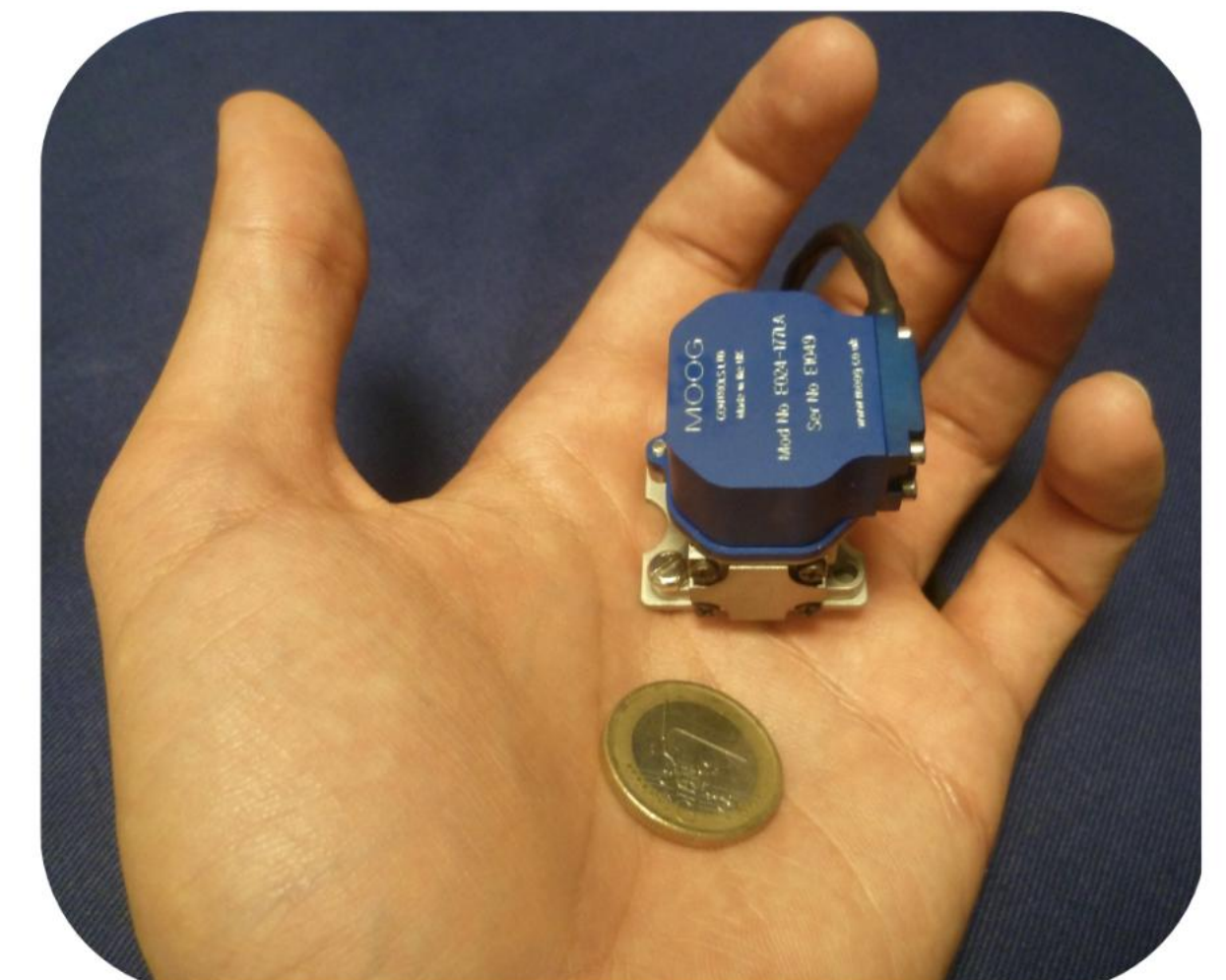
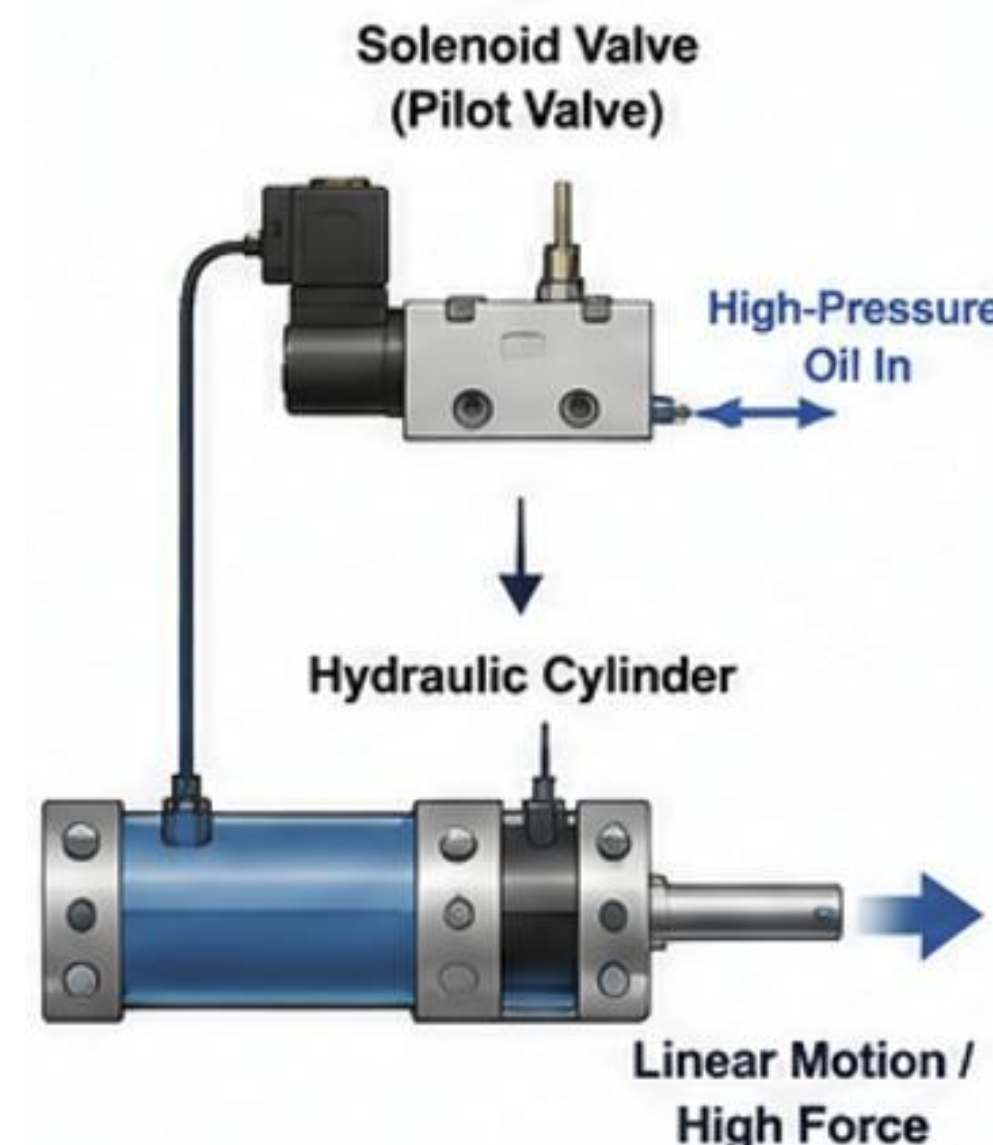
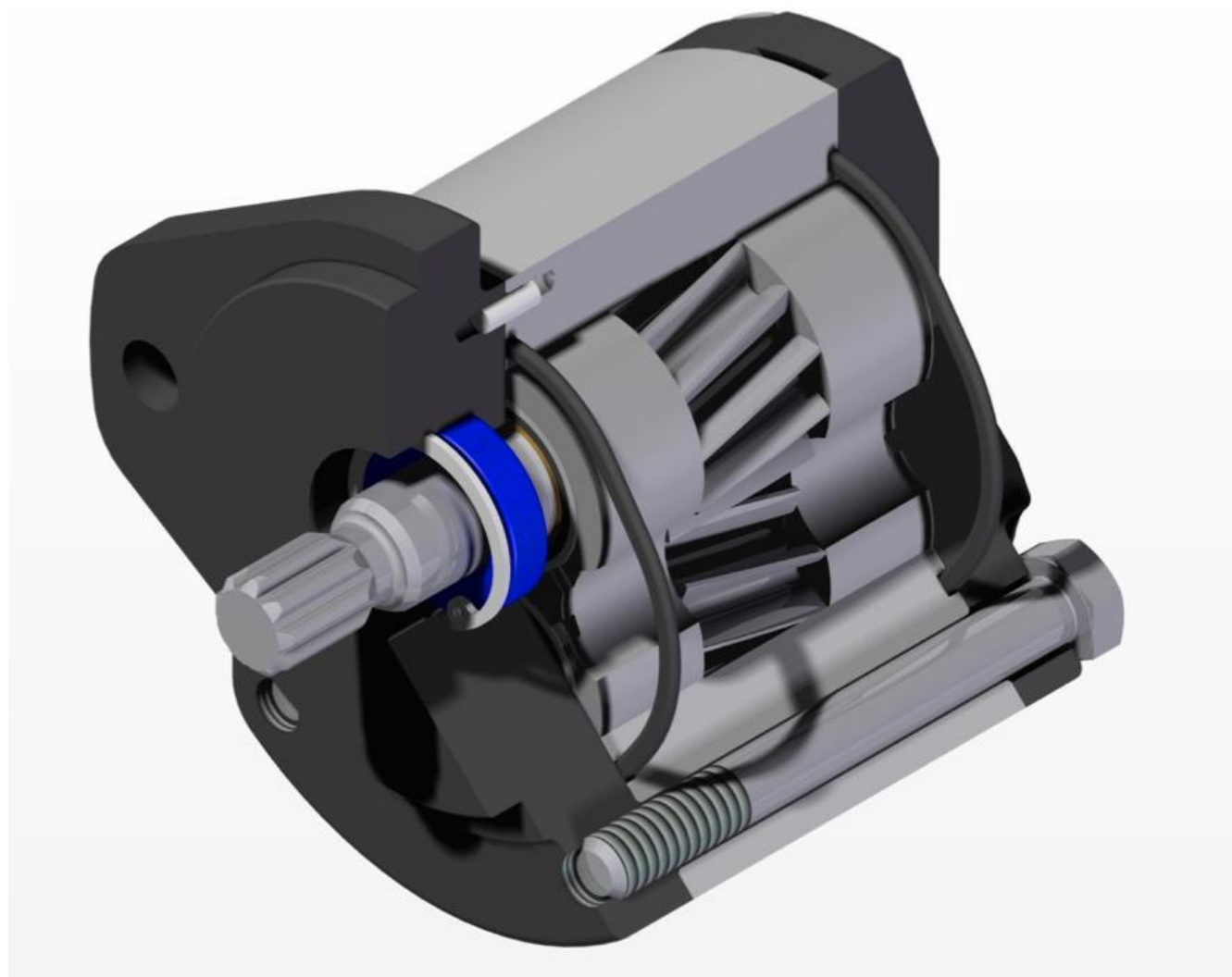
Type	Examples
Mechanical actuators	Lever, screw jack, hand crank
Electrical actuators	Thyristor, biopolar transistor, diode
Electromechanical actuators	AC motor, DC motor, step motor
Electromagnetic actuators	Electromagnet, linear solenoid
Hydraulic and pneumatic actuators	Hydraulic cylinder, pneumatic cylinder, piston, pressure control valves, air motors
Smart material actuators (includes thermal and magnetic actuators)	Shape memory alloy (SMA), ion exchange fluid, magnetostrictive material, bimetallic strip, piezoelectric bimorph
Micro- and nanoactuators	Electrostatic motor, microvalve, comb drive

Classification of Actuators by Energy Source

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1. Hydraulic: These actuators harness hydraulic power to perform mechanical functions and operations.

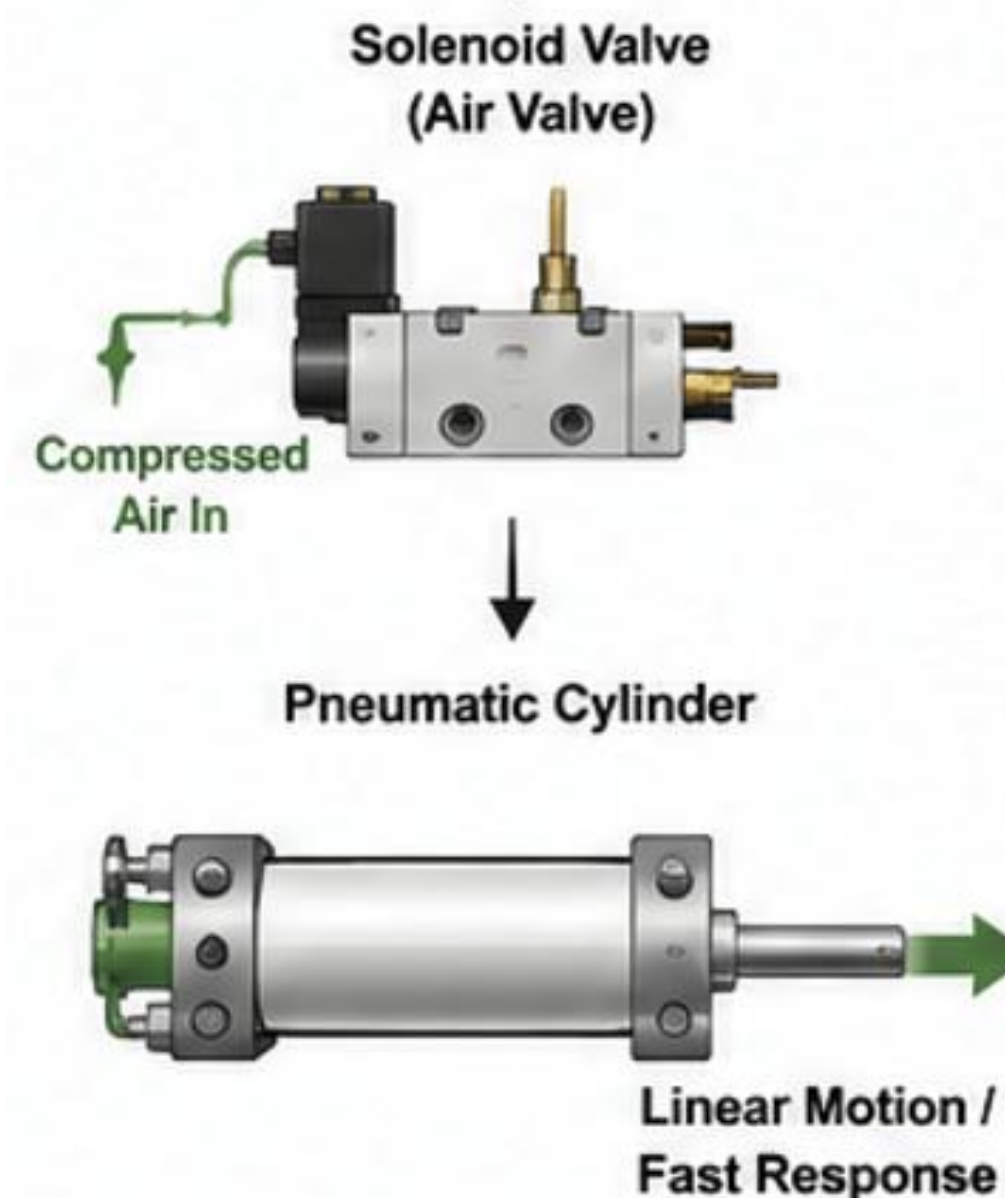


(a) Miniature MOOG E024 servovalves..

Classification of Actuators by Energy Source



2. Pneumatic: They are powered by a vacuum or compressed air at high pressure to implement the required type of motion.



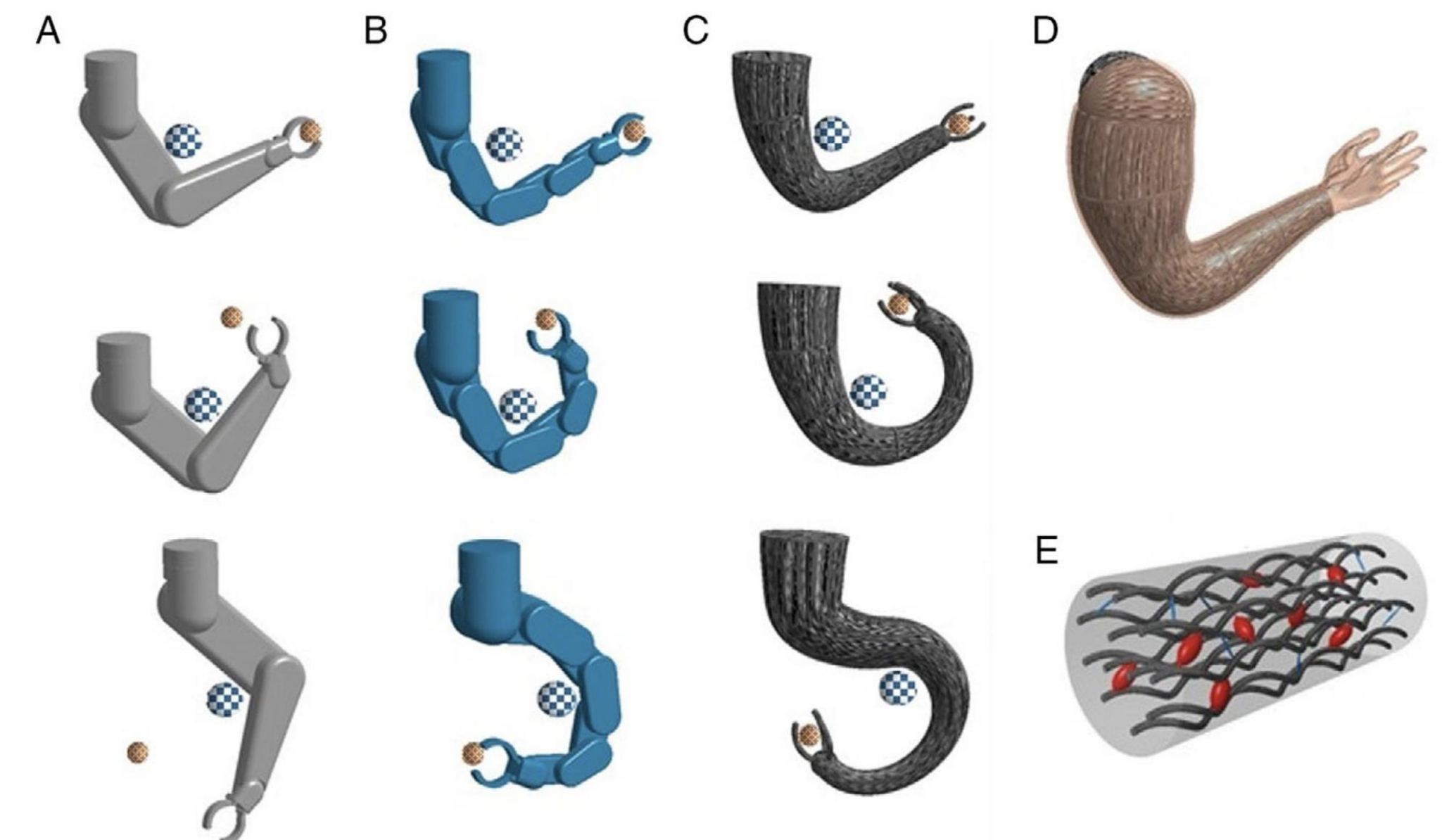
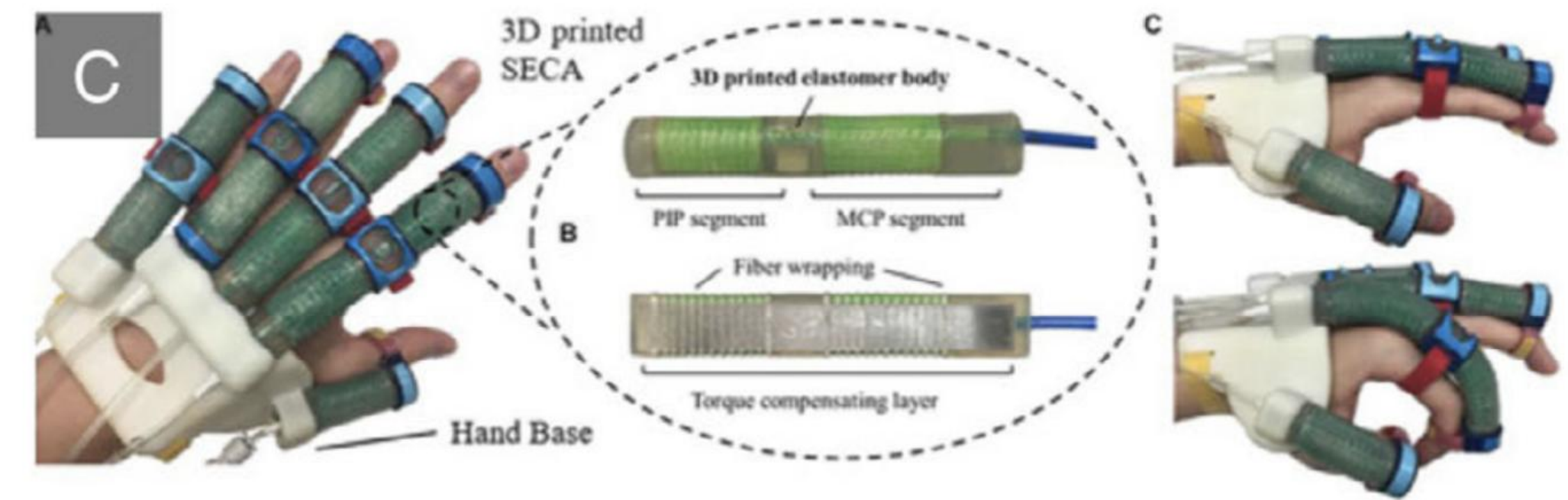
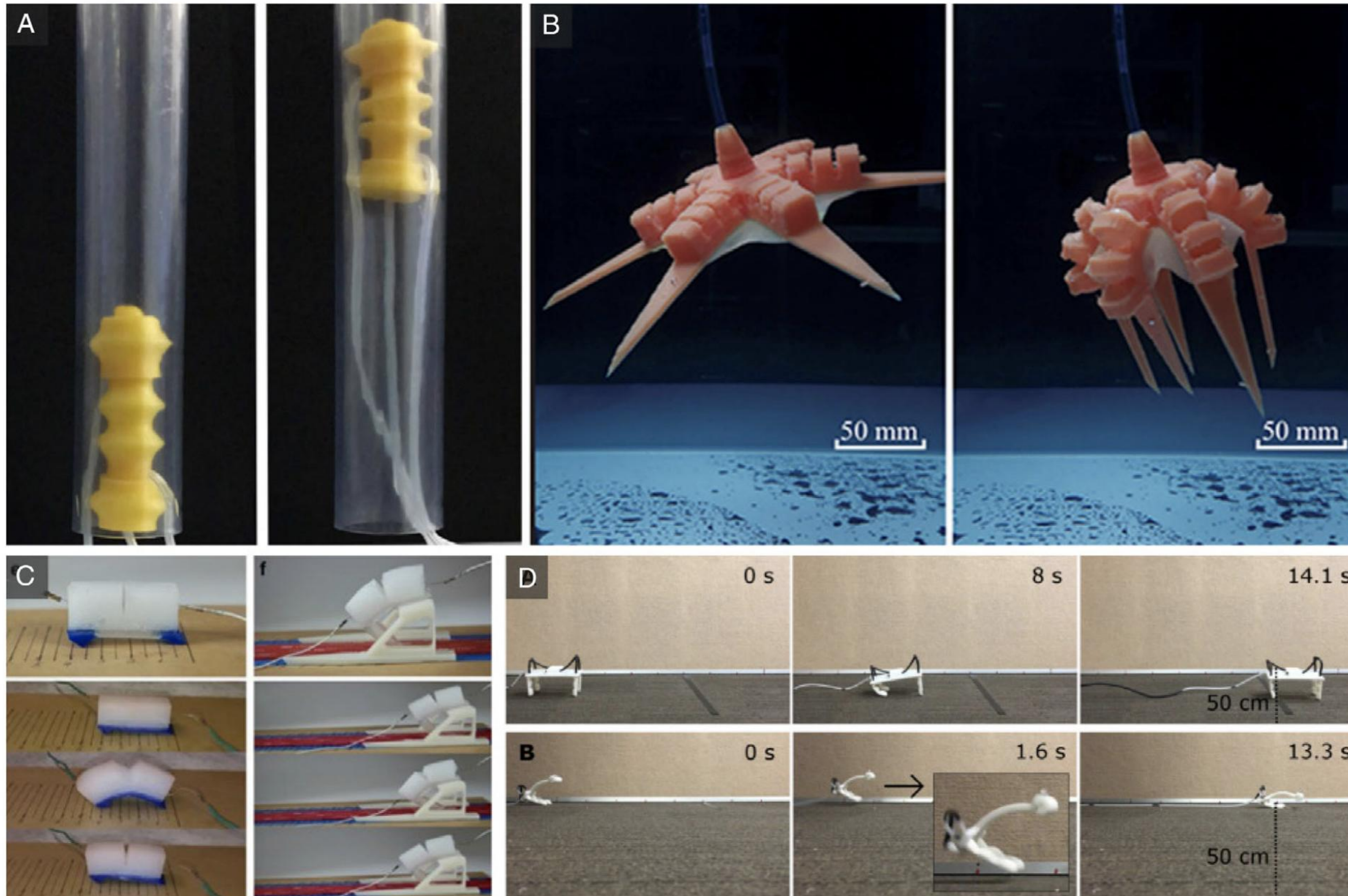
Example:

Powers automated assembly, packaging, sorting, clamping systems.



Soft Actuators

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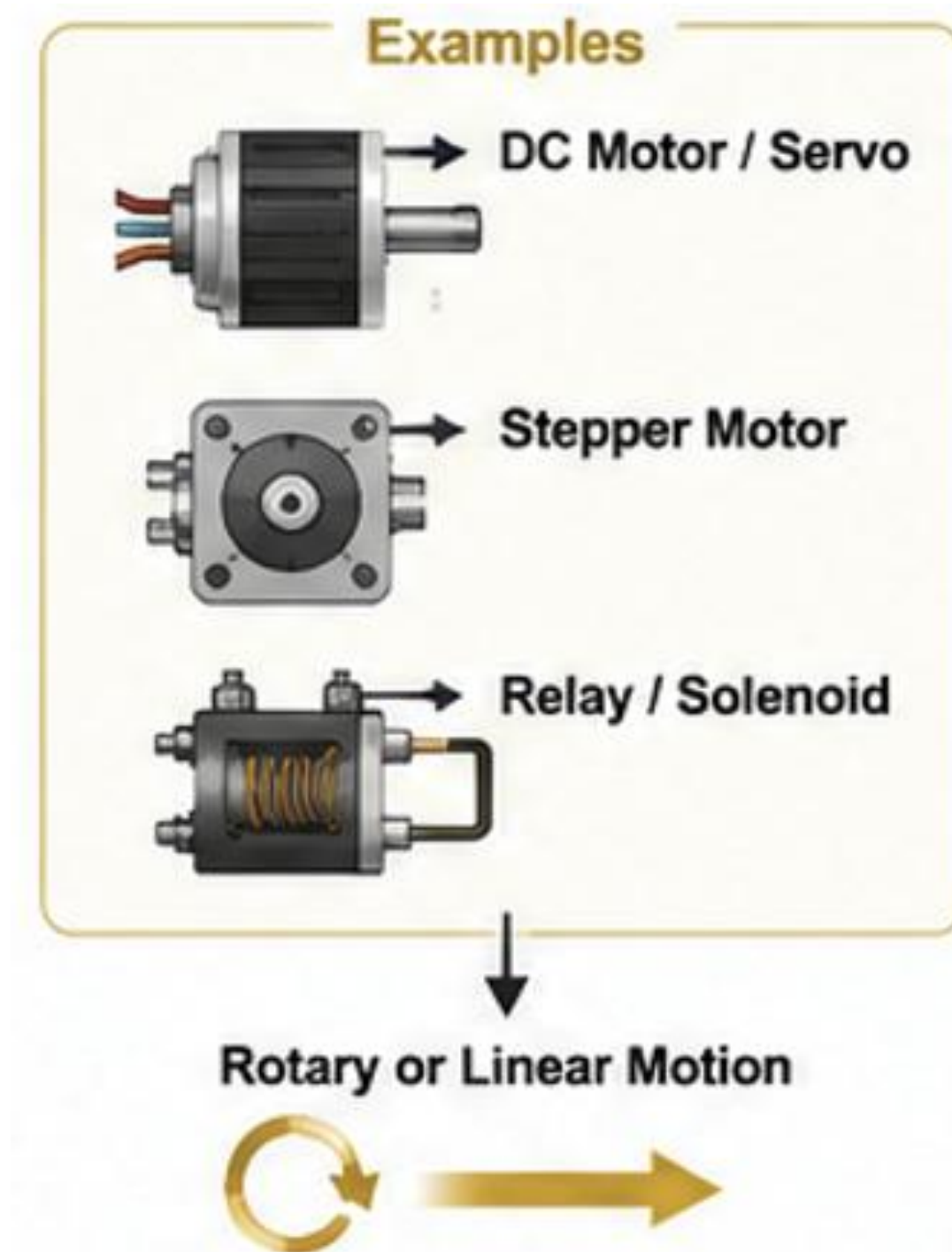
Tawk et al, A Review of 3D-Printable Soft Pneumatic Actuators and Sensors: Research Challenges and Opportunities

Classification of Actuators by Energy Source



3. Electrical: These actuators are heavily used in industrial settings to automate mechanical operations.

All other actuators mainly rely in electrical energy as an intermediate energy conversion.



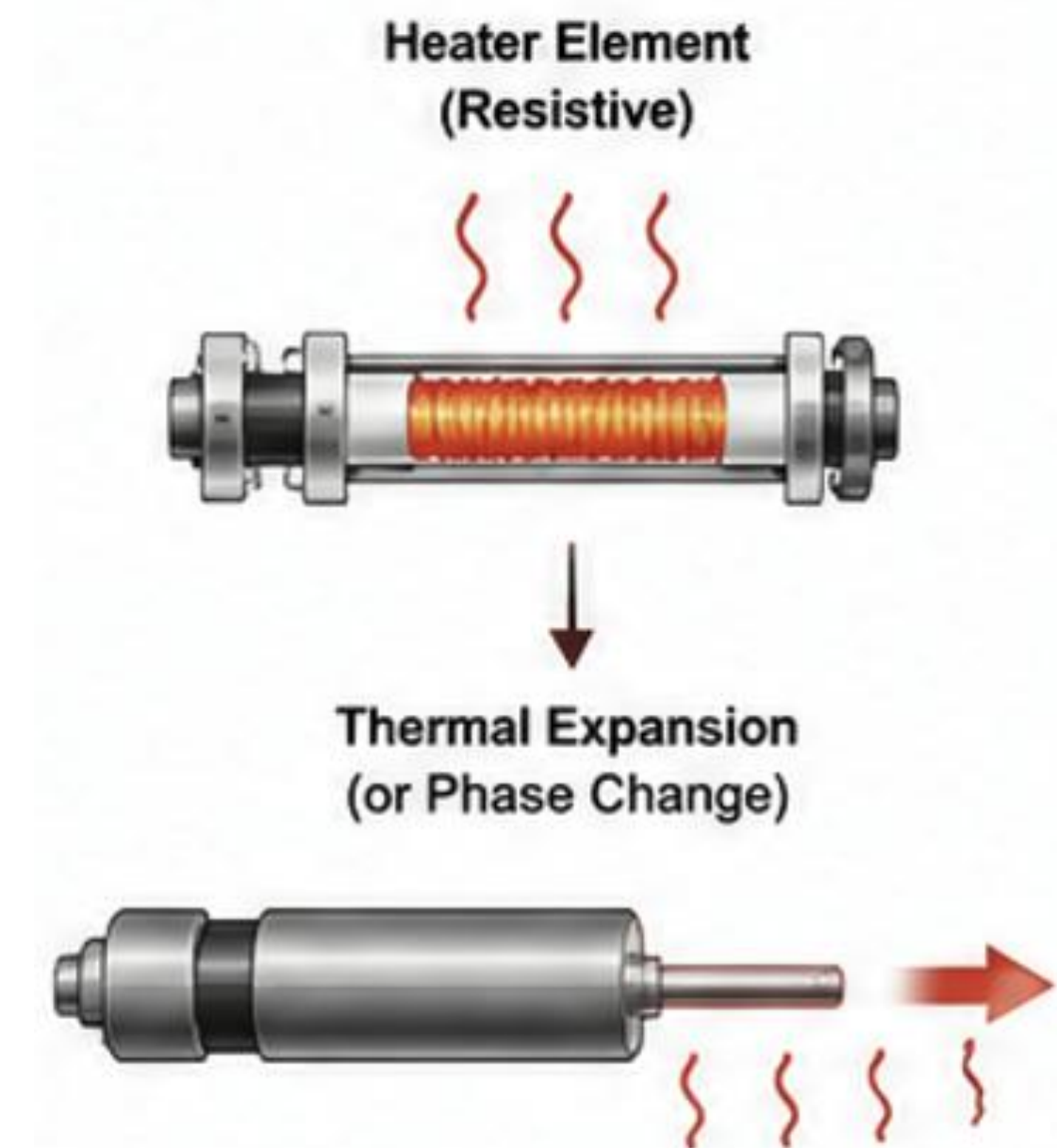
Classification of Actuators by Energy Source



4. Thermal: The thermal actuators have thermal-sensitive material fitted inside, which is used to produce linear motion.

The word thermal implies that these actuators are used in response to temperature changes.

The most popular use case includes shutting off valves and operating latches or switches.





Bimetales

<https://www.youtube.com/shorts/RAEMm63guVY>



SMA o Shape Memory Alloys (Aleaciones con Memoria de Forma)

<https://www.youtube.com/watch?v=jdNyZs8JI3I>

Classification of Actuators by Energy Source



5. Magnetic: These types of actuators convert electromagnetic energy into mechanical output and operate in a linear or rotary direction.

Magnetic actuators can provide continuous mechanical operation and are popularly used in the automotive and aerospace industries.

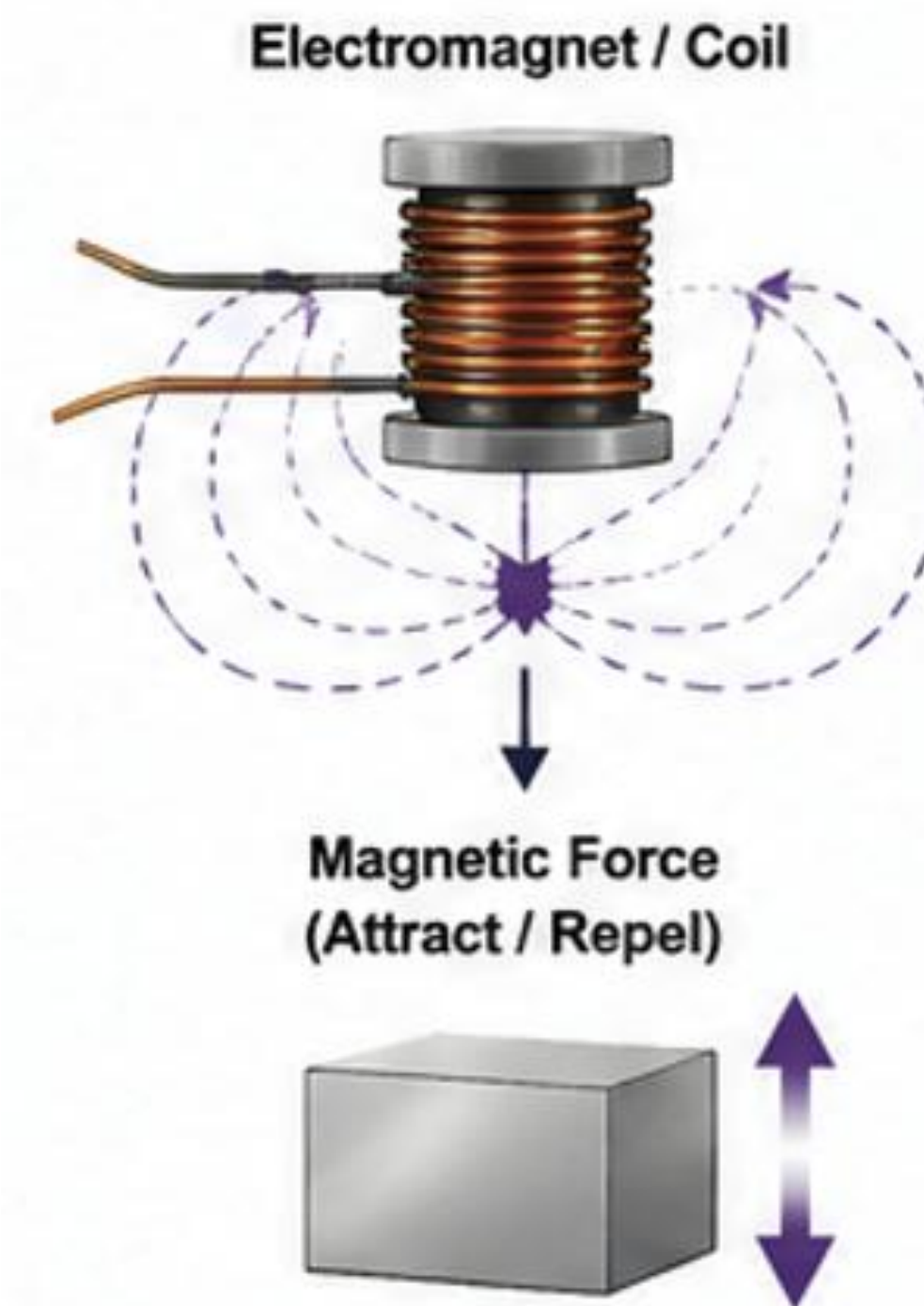
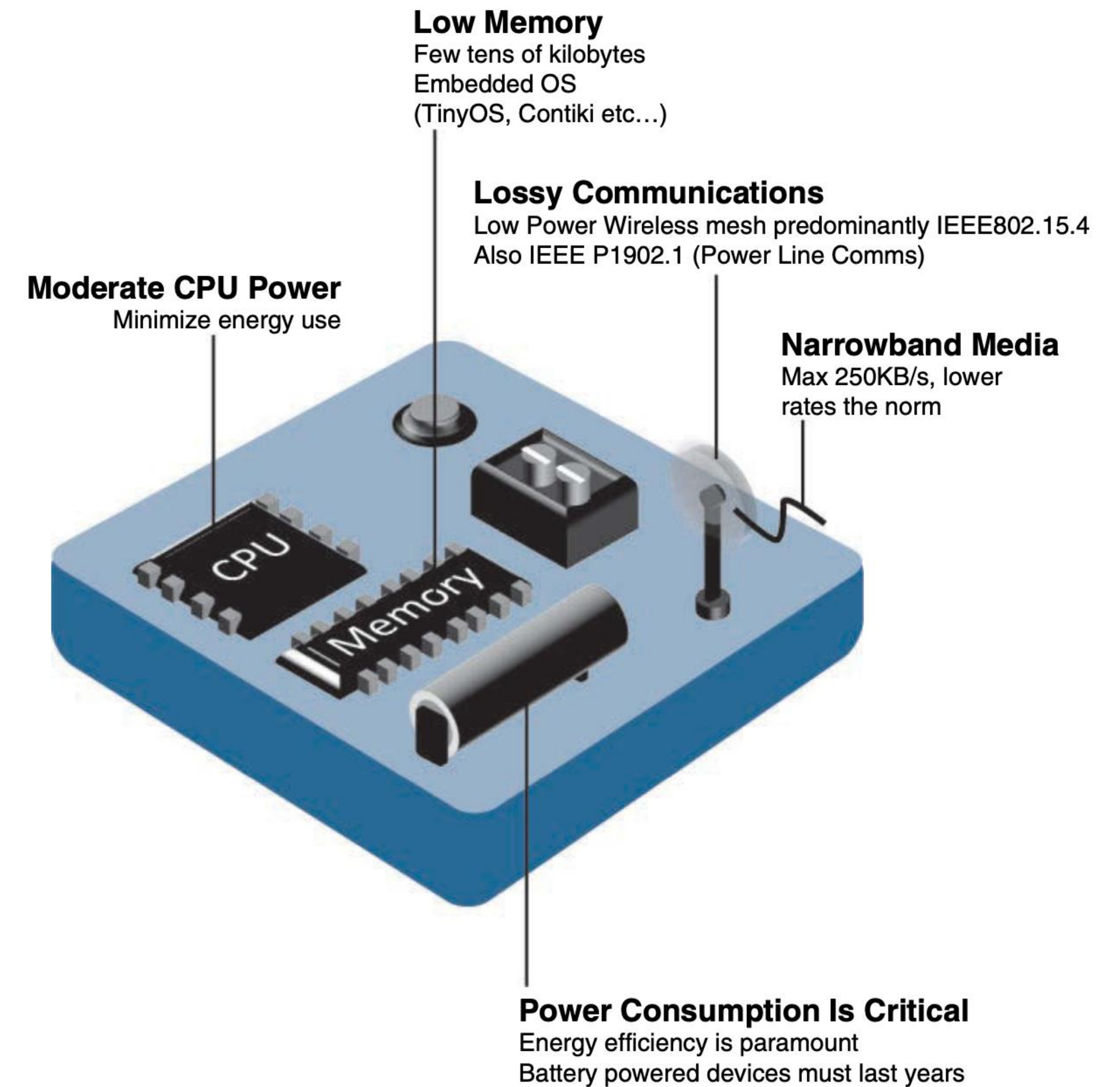
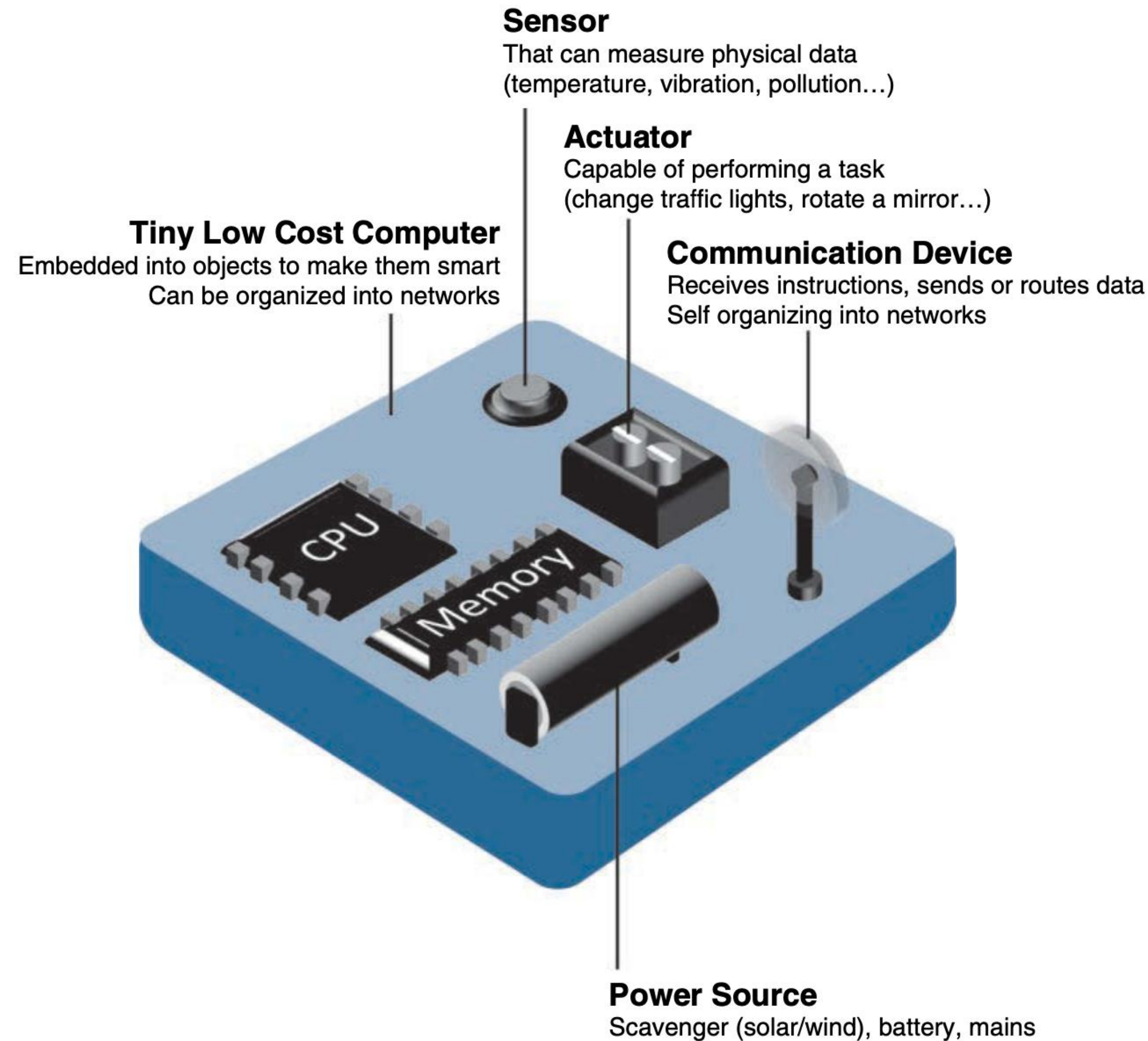


Figure 6. Magnetic float sensor [100], Courtesy of t

Smart sensor: E.g. Arduino boards

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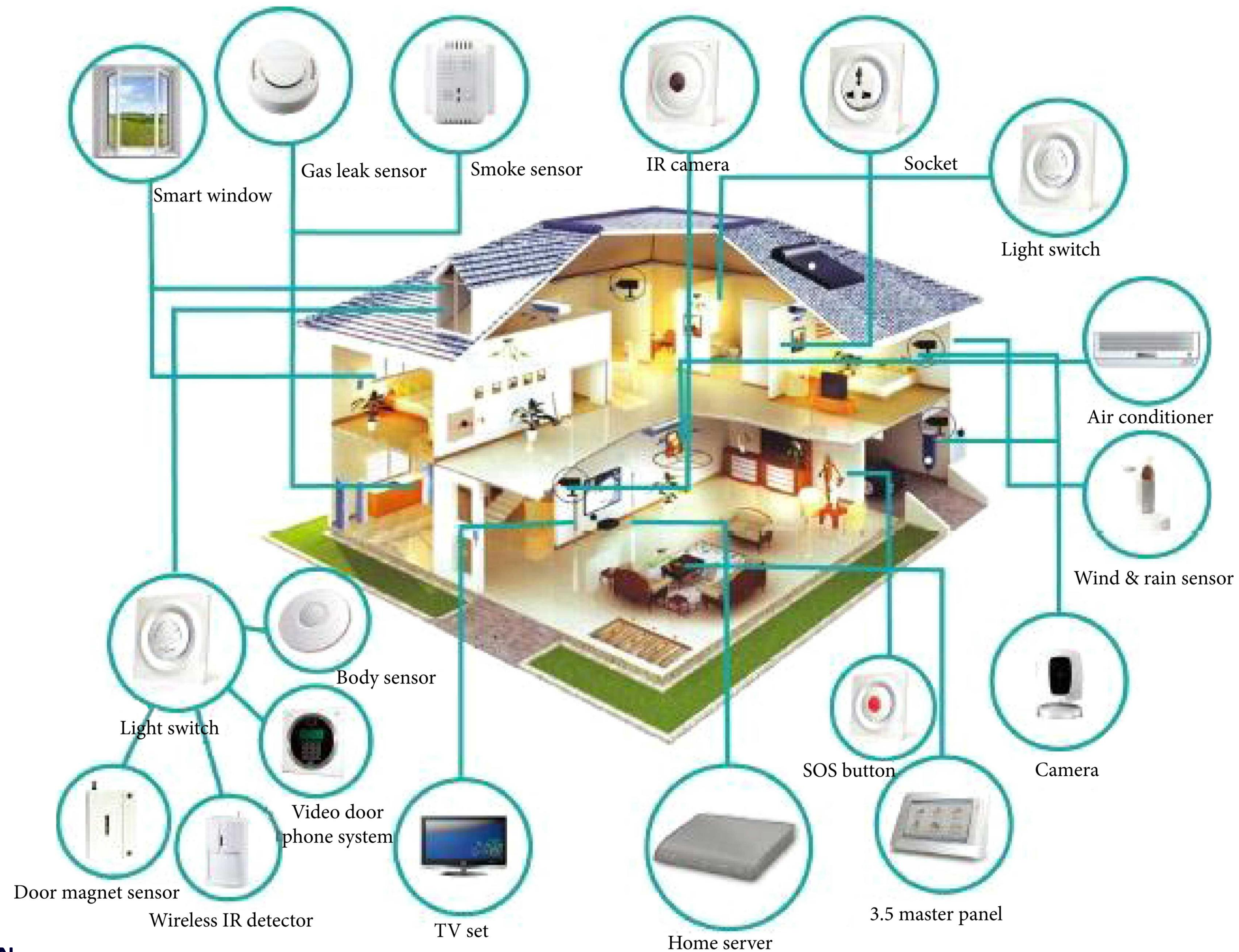
Case Study

Wheatstone bridge

Conclusions

Actuators: Case study 1 - Smarthome

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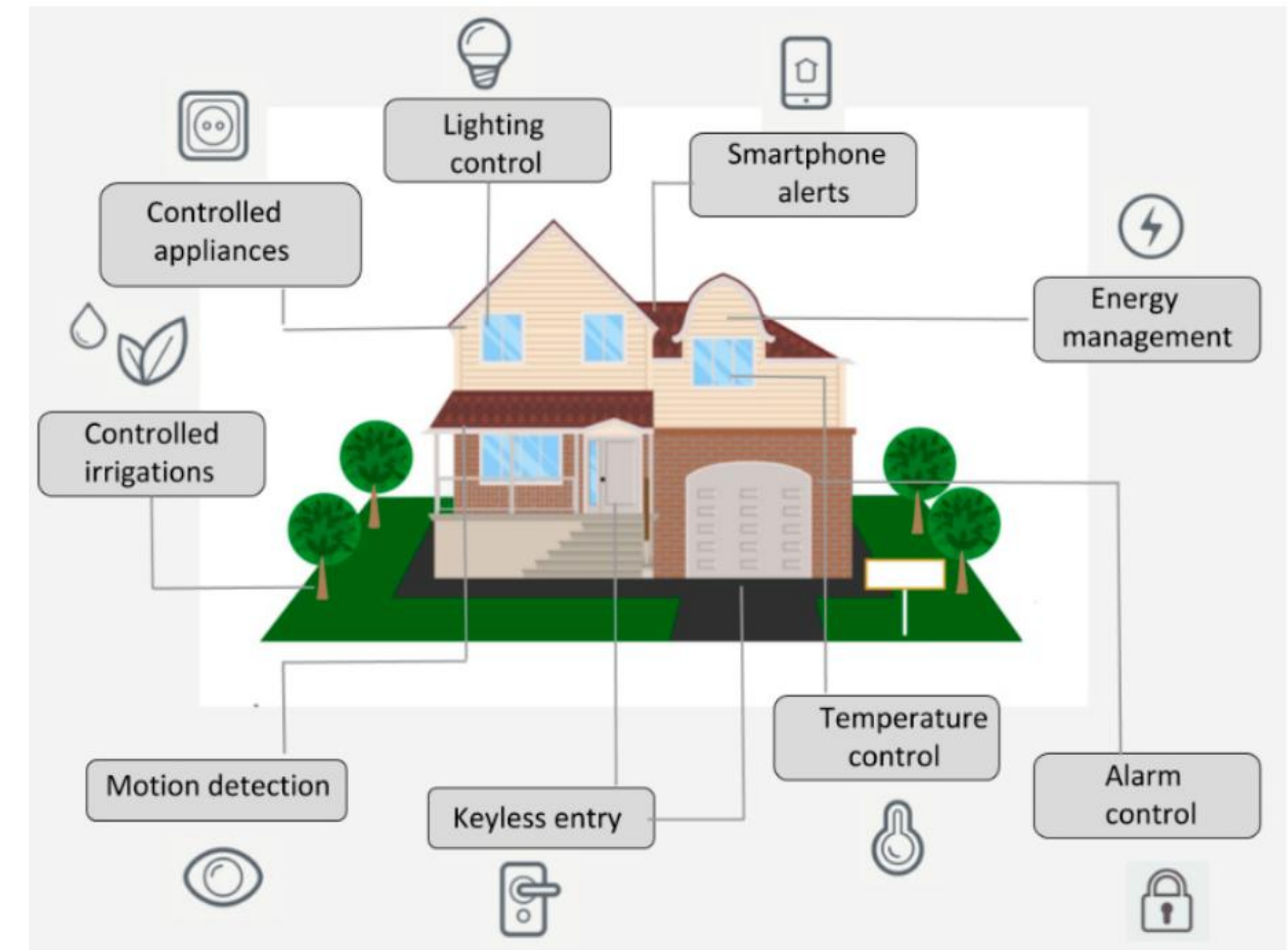


Yang et. Al. "Advanced Internet of Things and Big Data Technology for Smart Human-Care Services"

Actuators: Case study 1 - Smarthome

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Home Automation System	Communication	Controller	User Interface	Applications
[4]	Bluetooth	PIC	mobile app	control indoor appliances
[5]	Bluetooth	Arduino	mobile app	control appliances indoor and outdoor, within short range
[6]	Bluetooth, GSM	PIC	mobile app	control appliances indoor and outdoor
[7]	ZigBee, Ethernet	Arduino MEGA	mobile app	control appliances indoor
[8]	X10, Serial, EIB, ZigBee, Bluetooth,	32-bit ARM microcontroller	Control panel (touch pad), desktop based	indoor automation solution
[9]	Wi-Fi, ZigBee	Raspberry PI, NodeMCU		controlling humidity, temperature, luminosity, movement, and current
[10]	ZigBee	Laptop/PC server	mobile app	control of indoor appliances but not actually implemented
[11]	ZigBee, Wi-Fi	Linux board	GUI interface	control HVAC appliances
[12]	ZigBee, Wi-Fi, Ethernet	Raspberry PI	web-based, mobile app	remote control of appliances (IP cams, smart plugs)
[13]	Wi-Fi	TI-CC3200 MCU	mobile app	control indoor appliances, monitor the soil moisture
[14]	Wi-Fi	NodeMCU	web-based	control indoor appliances
[15]	Bluetooth, Wi-Fi	Raspberry PI	mobile app	control indoor appliances
[16]	Wi-Fi	Arduino mega	web-based, mobile app	control of indoor appliances
[17]	Wi-Fi	PC server	web-based, mobile app	security, energy management
[18]	Wi-Fi, IR	PC server	mobile app	control of indoor appliances
[19]	Wi-Fi	Arduino	mobile app	control indoor appliances, video surveillance
[20]	Bluetooth	Arduino	mobile app	control indoor appliances, energy management
[21]	Wi-Fi	Arduino, ESP8266	mobile app	control indoor appliances



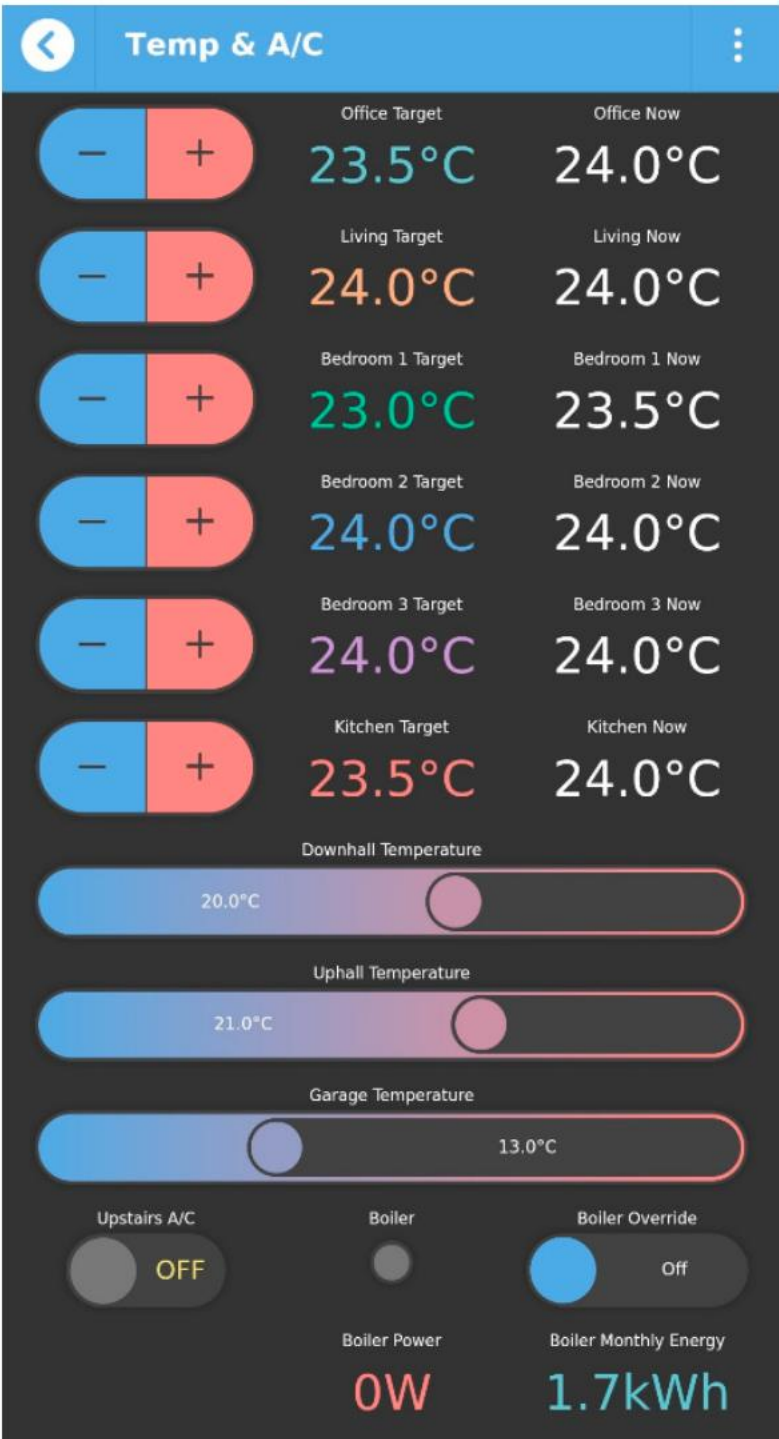
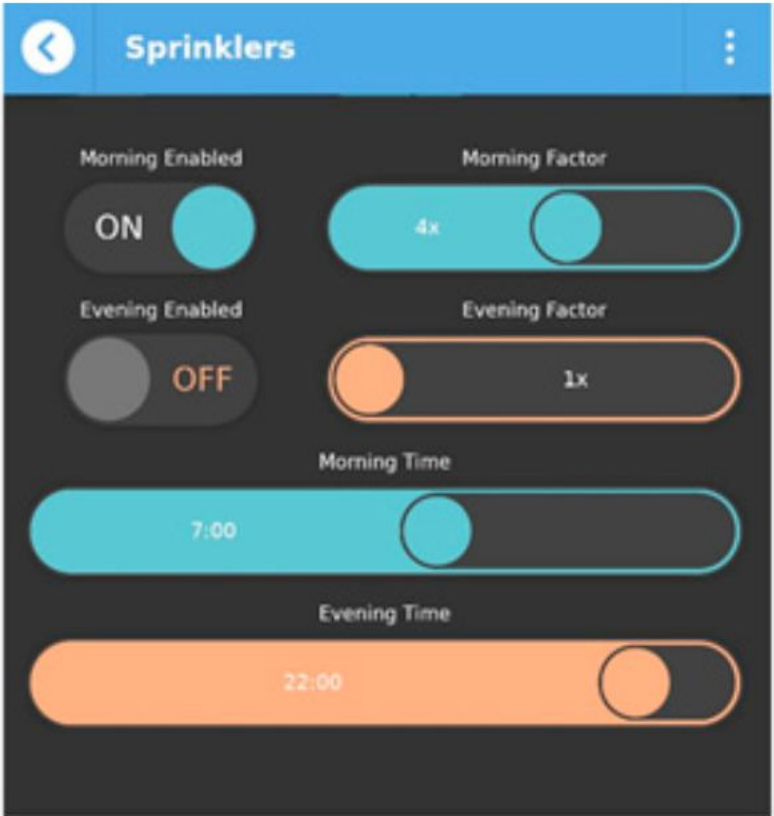
Features comparison for home automation system published in the last 10 years.

Actuators: Case study 1 - Smarthome

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Comparison of the most relevant open-source home automation platforms.

System	Development Language	API	Other Features
OpenHAB	Java	Representational state transfer (REST)	web interface, many protocols, many plugins, MQTT, EPL v1 license, extensive documentation
HomeAssistant	Python	REST/Python/Websocket APIs	web interface, many protocols, many plugins, MQTT, Apache 2.0 license, extensive documentation
Domoticz	C++	JSON based	web interface, many protocols, many plugins, MQTT, GPL v3 license, extensive documentation
Calaos	C++	JSON based	web interface, a few protocols, under development plugins, MQTT, GPL v3 license, extensive documentation (in French)
Jeedom	PHP	JSON RPC and HTTP based	web interface, many protocols, many plugins, MQTT, GPL v2 license, extensive documentation (in French)
Fhem	Perl	ASCCII commands	web interface, many protocols, many plugins, MQTT, GPL v2 license, extensive documentation (in German)
qToogle	Python	JSON based REST	web interface, many protocols, a few plugins (undergoing continuous development), Hypertext Transfer Protocol (HTTP) based messaging, Apache 2.0 license, extensive documentation (in English on Github)



(b)



(c)

Actuators: Case study 2 – Precision agriculture – Actuators + sensors

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Oré et al., “Crop Growth Monitoring with Drone-Borne DInSAR “

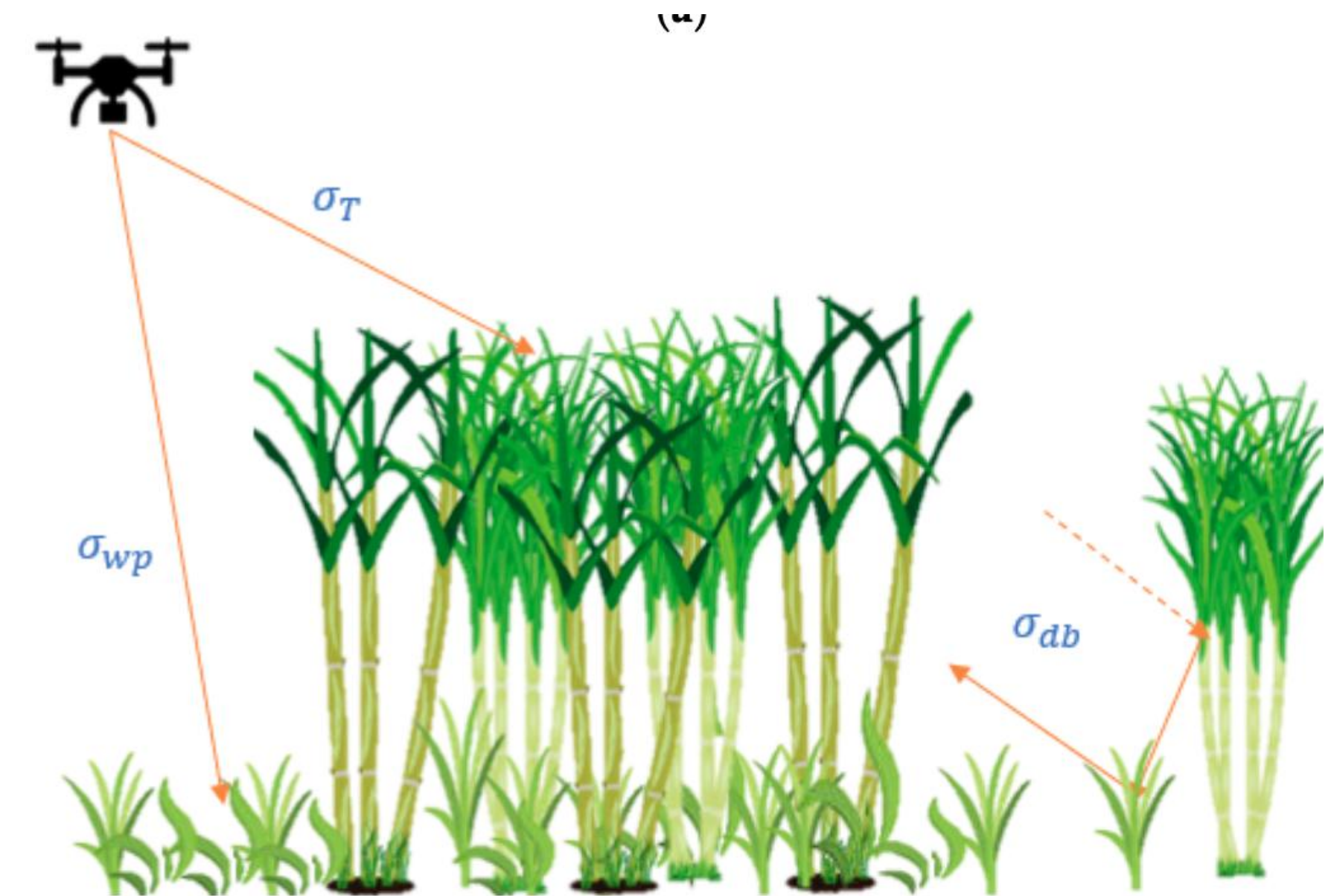
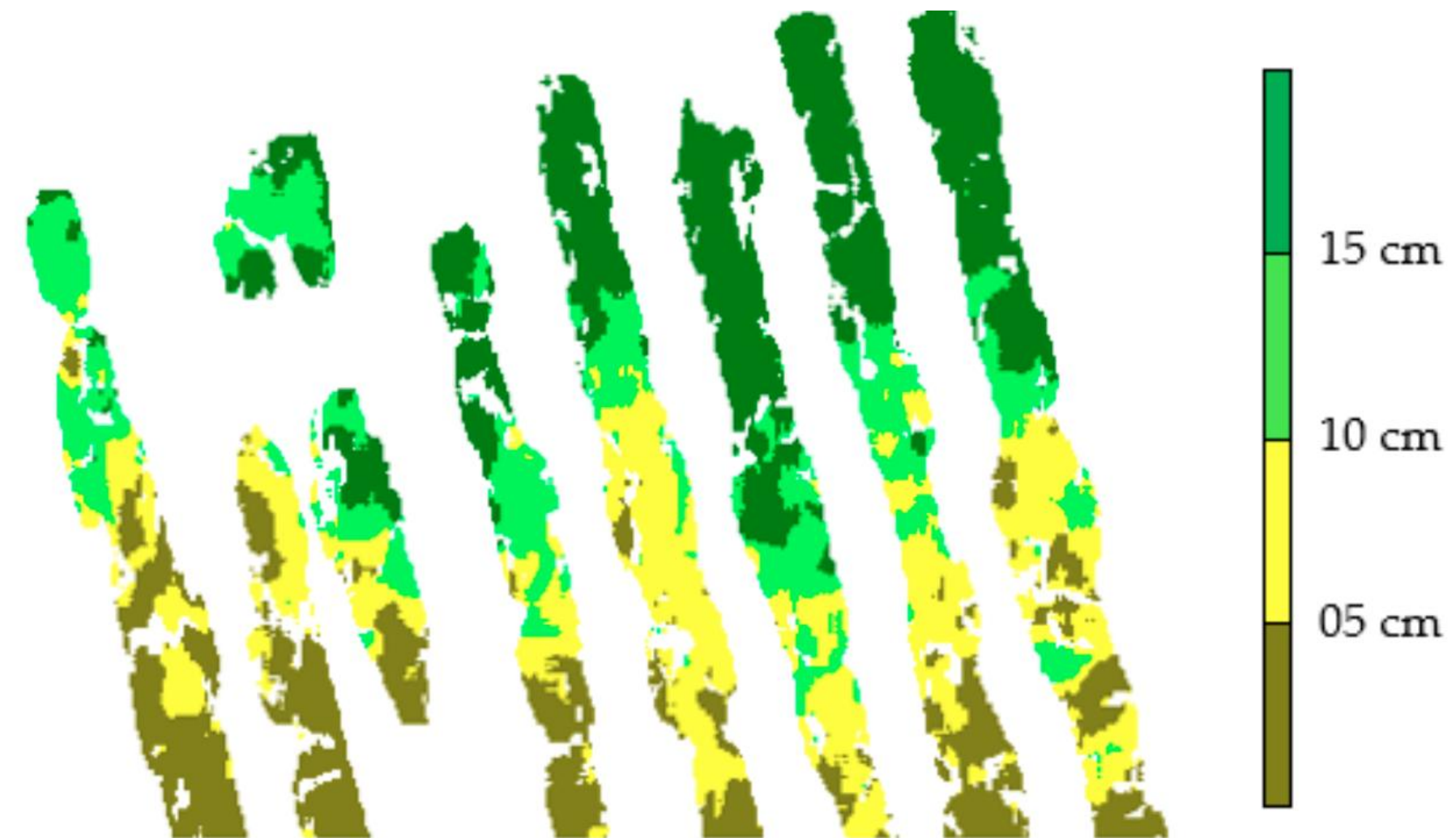
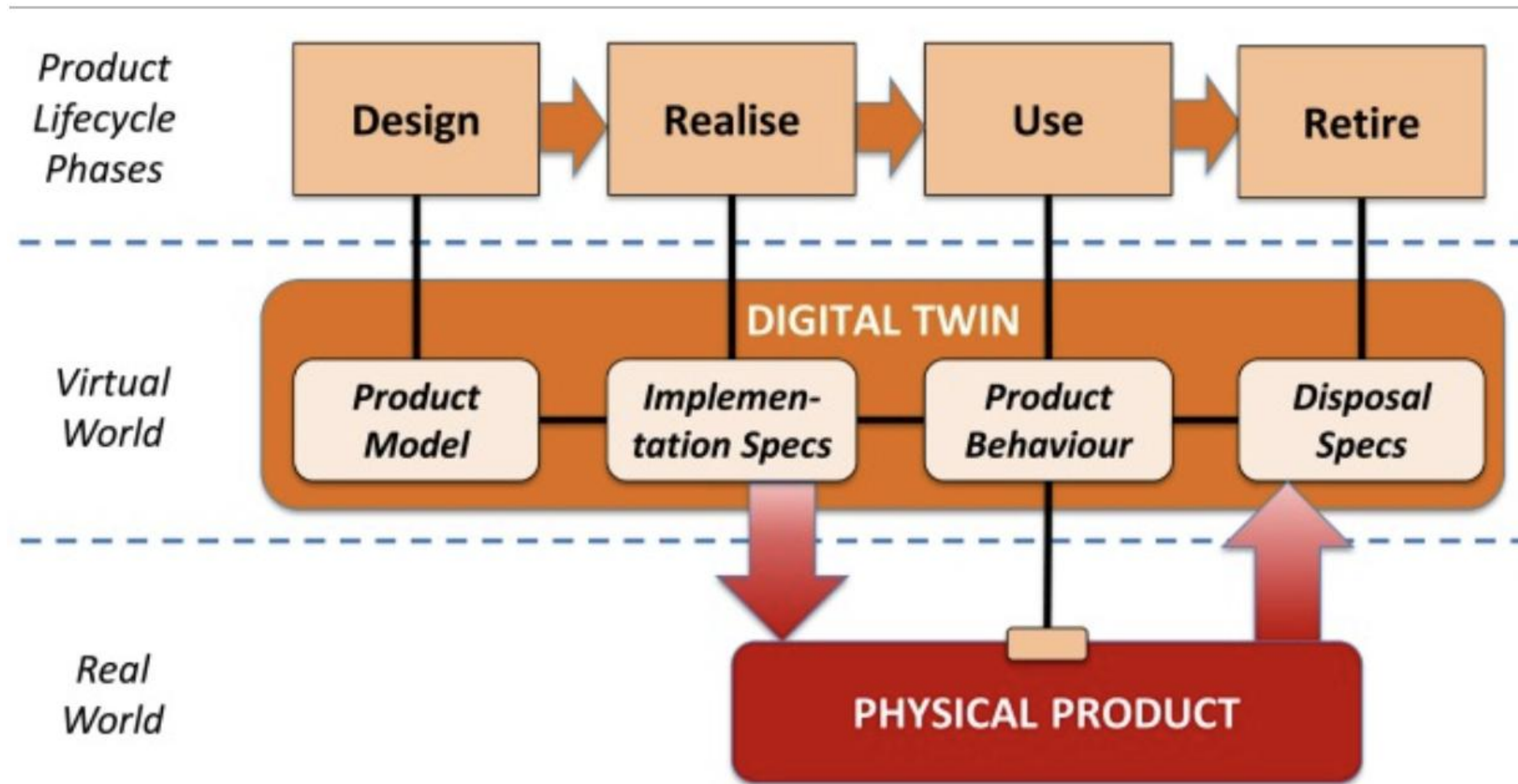


Figure 11. Growth map of the coffee crop from 11 December 2018, to 17 April 2019.

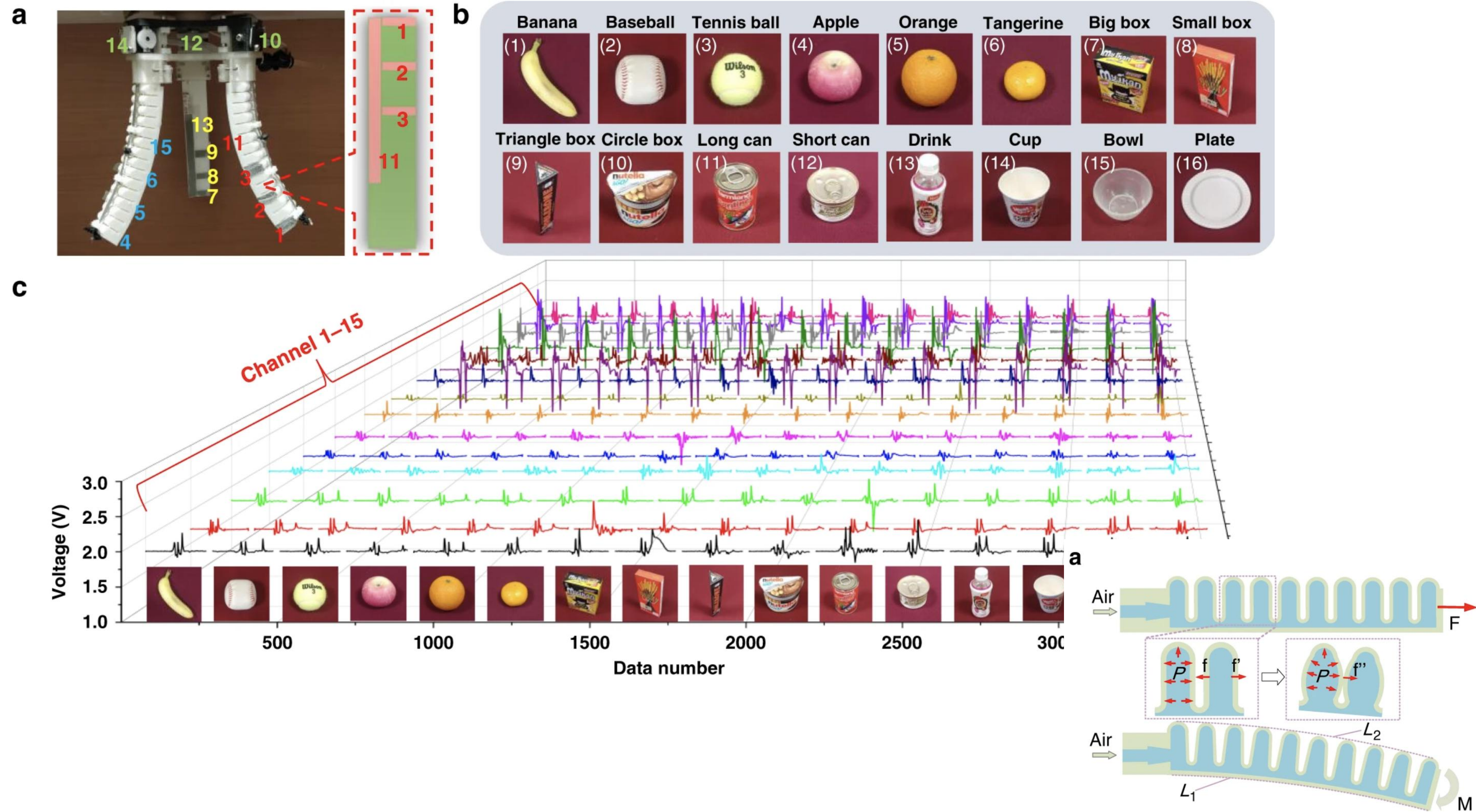
Actuators: Case study 3 - Digital Twin

“Verdouw et al., Digital twins in smart farming”



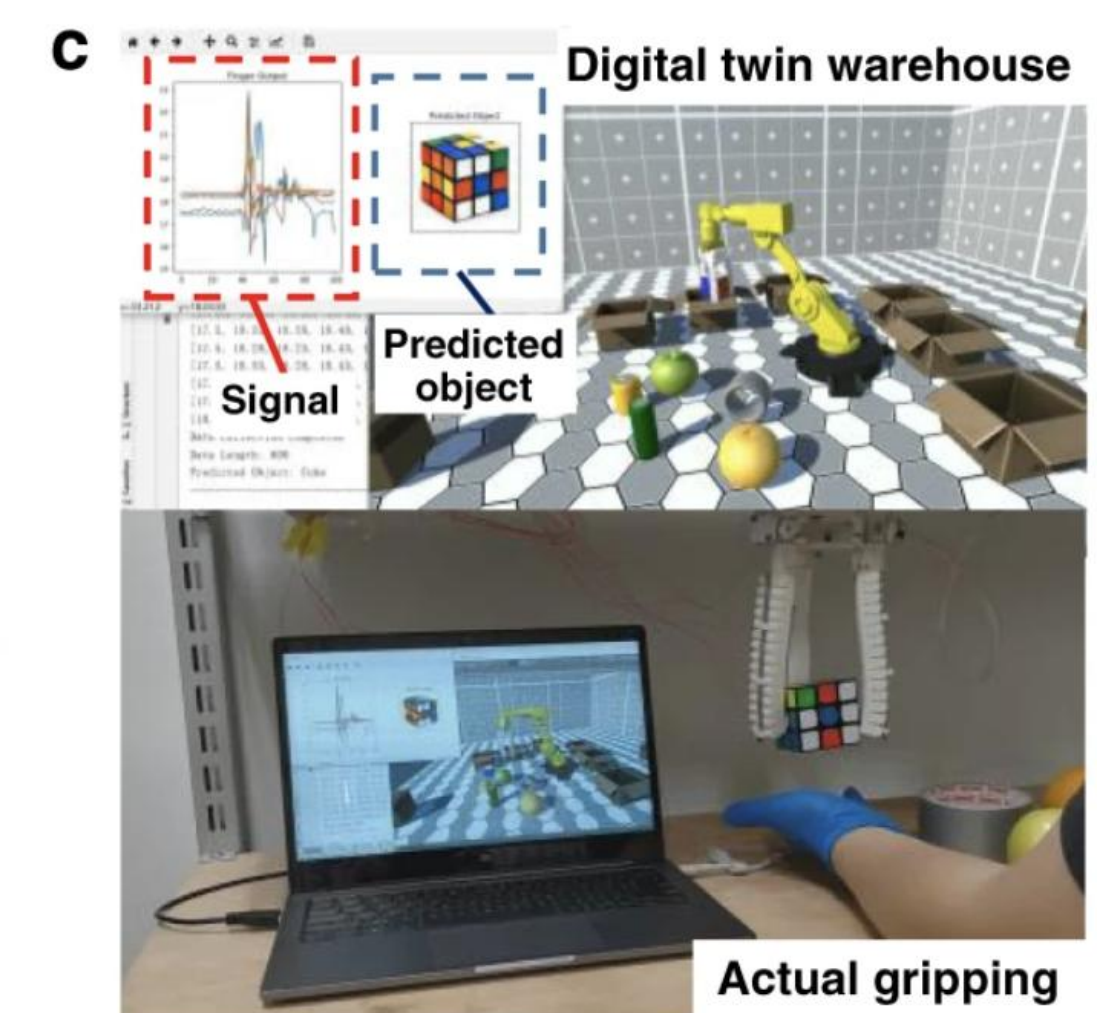
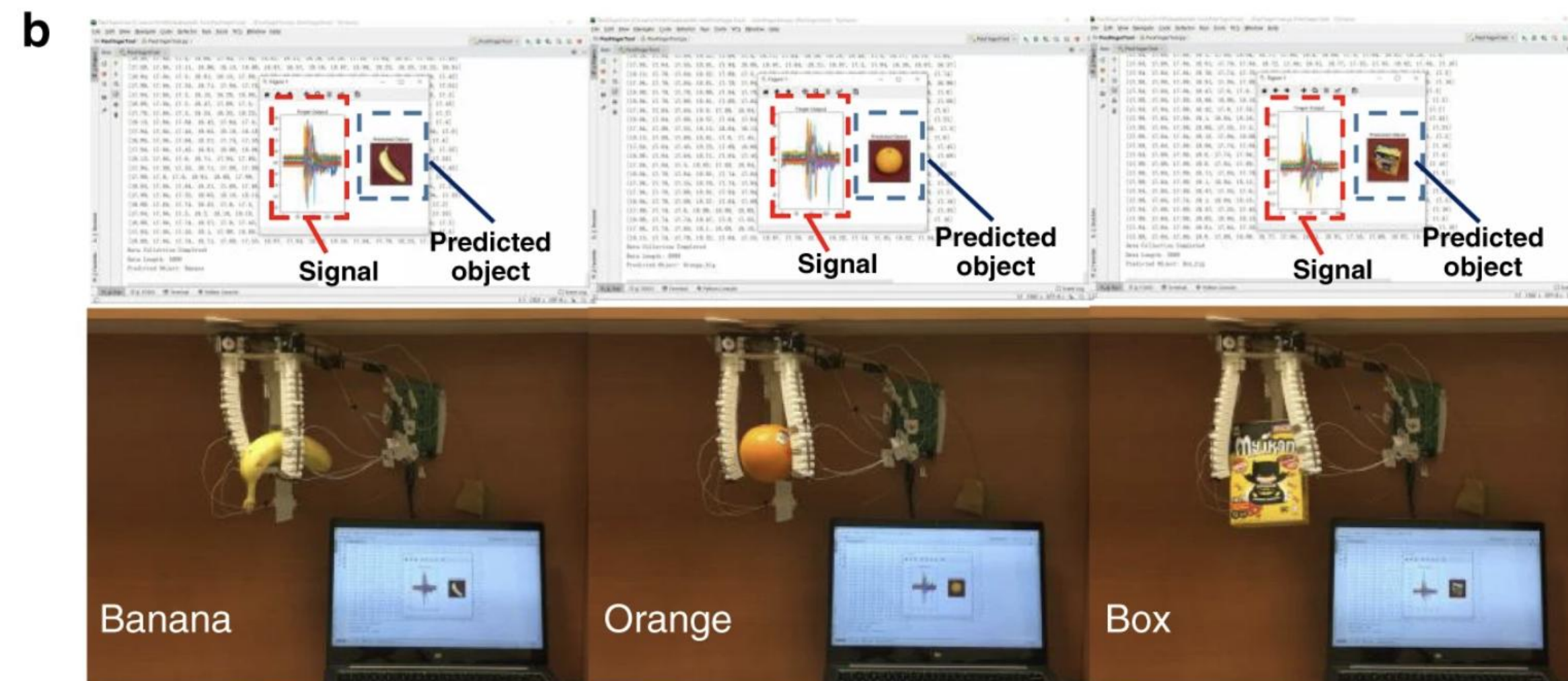
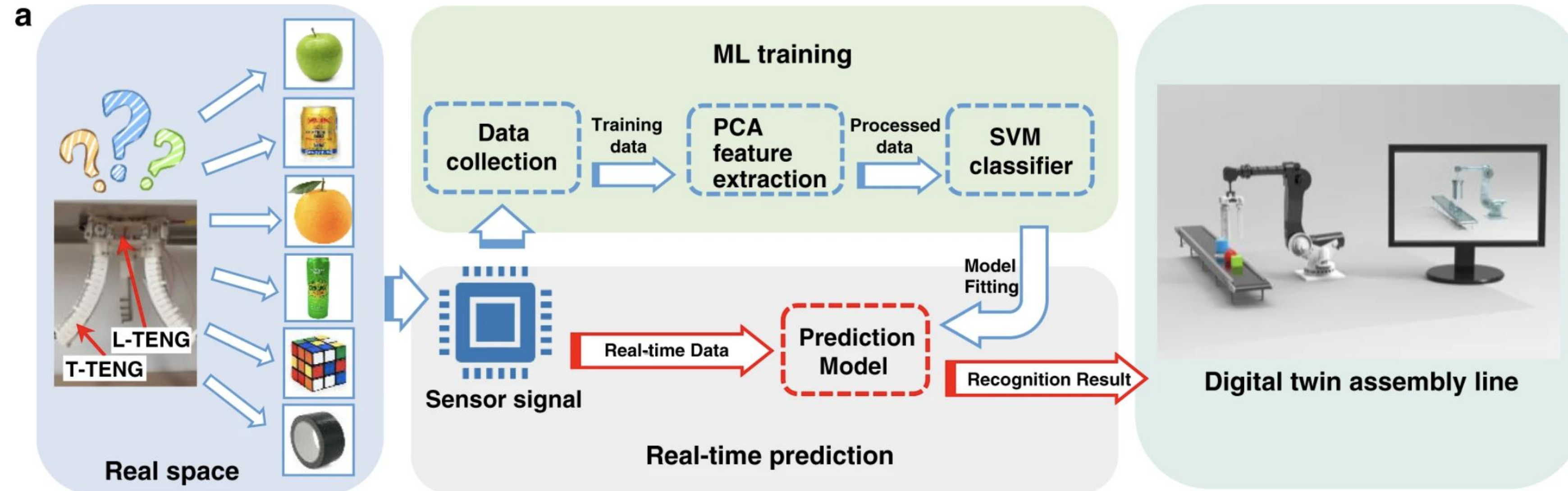
Actuators: Case study 3 - Digital Twin

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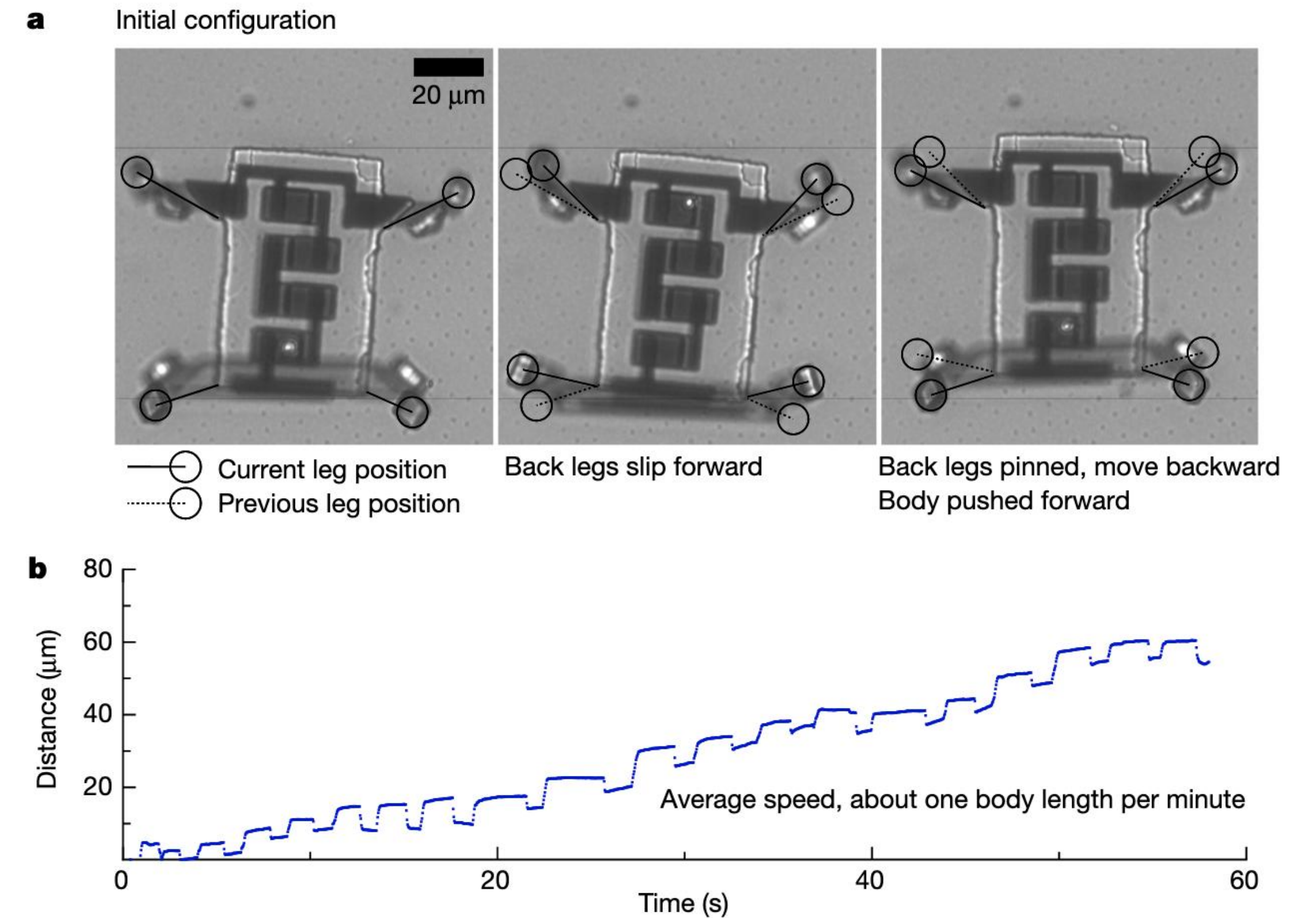
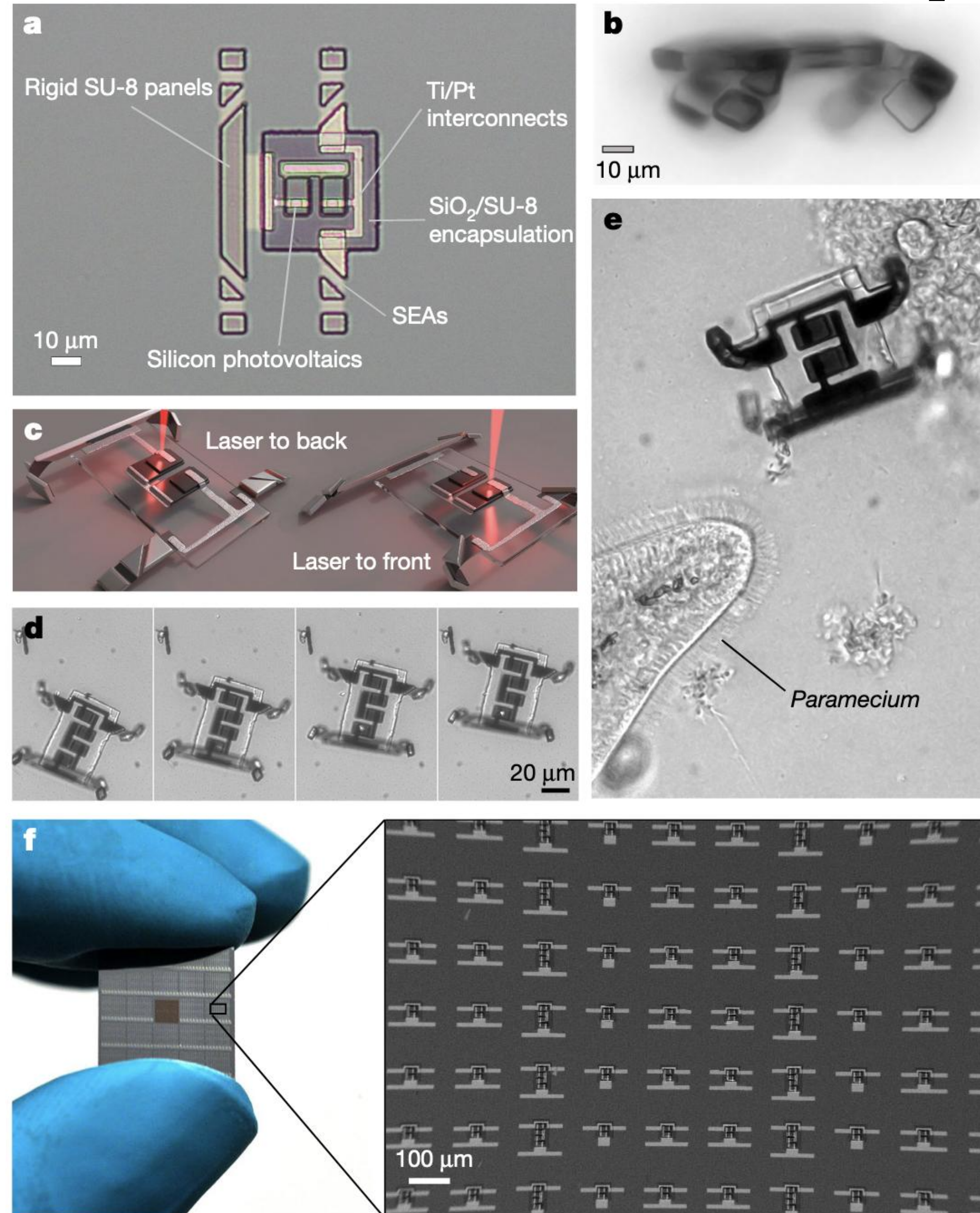
Actuators: Case study 3 - Digital Twin

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Actuators: Case study 4 – Microscopic robots

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Miskin et al., “Electronically integrated, mass-manufactured, microscopic robots”

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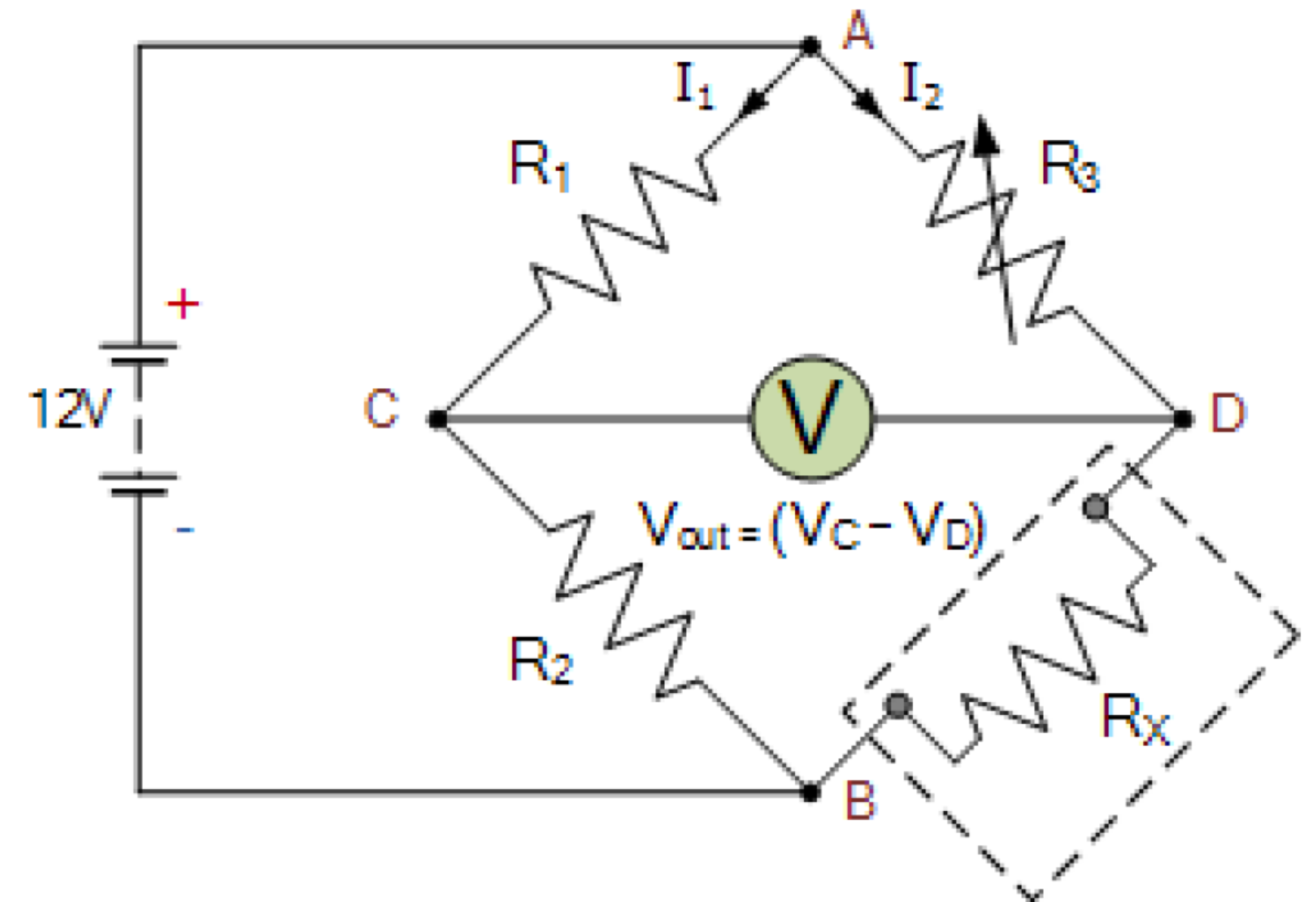
Wheatstone bridge

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El puente de Wheatstone es un circuito eléctrico en forma de rombo compuesto por cuatro resistencias conectadas en serie-paralelo.

Se utiliza principalmente para medir resistencias eléctricas desconocidas con alta precisión.

Es ampliamente empleado en instrumentación, especialmente en sensores basados en variaciones resistivas (galgas extensométricas, termistores, etc.).



Wheatstone bridge

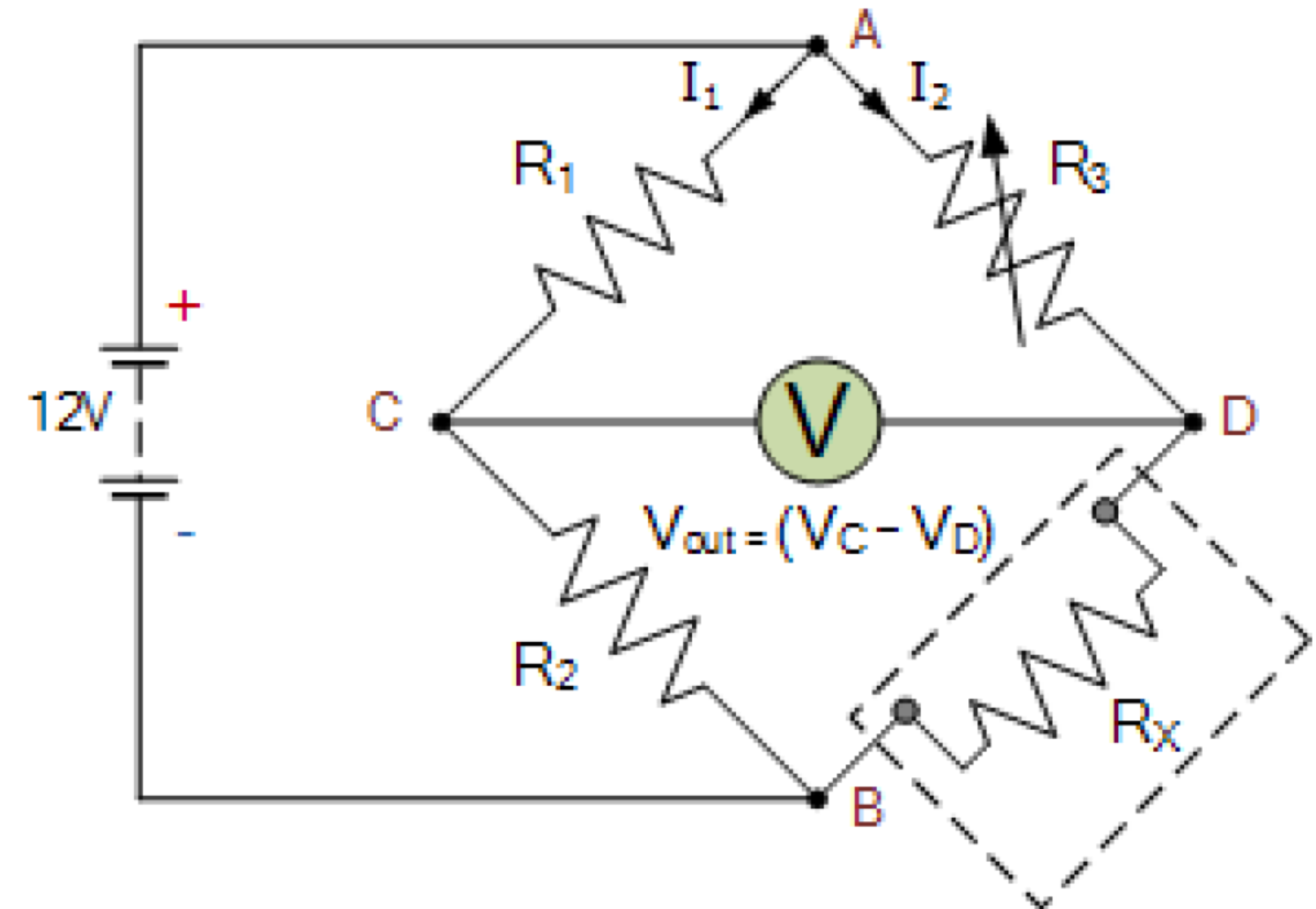
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Cuando la relación de resistencias cumple la condición de equilibrio:

$$\frac{R_1}{R_2} = \frac{R_3}{R_x}$$

el voltaje entre los terminales del detector es cero.

A partir de esta condición se puede calcular el valor de la resistencia desconocida R_x .



Wheatstone bridge

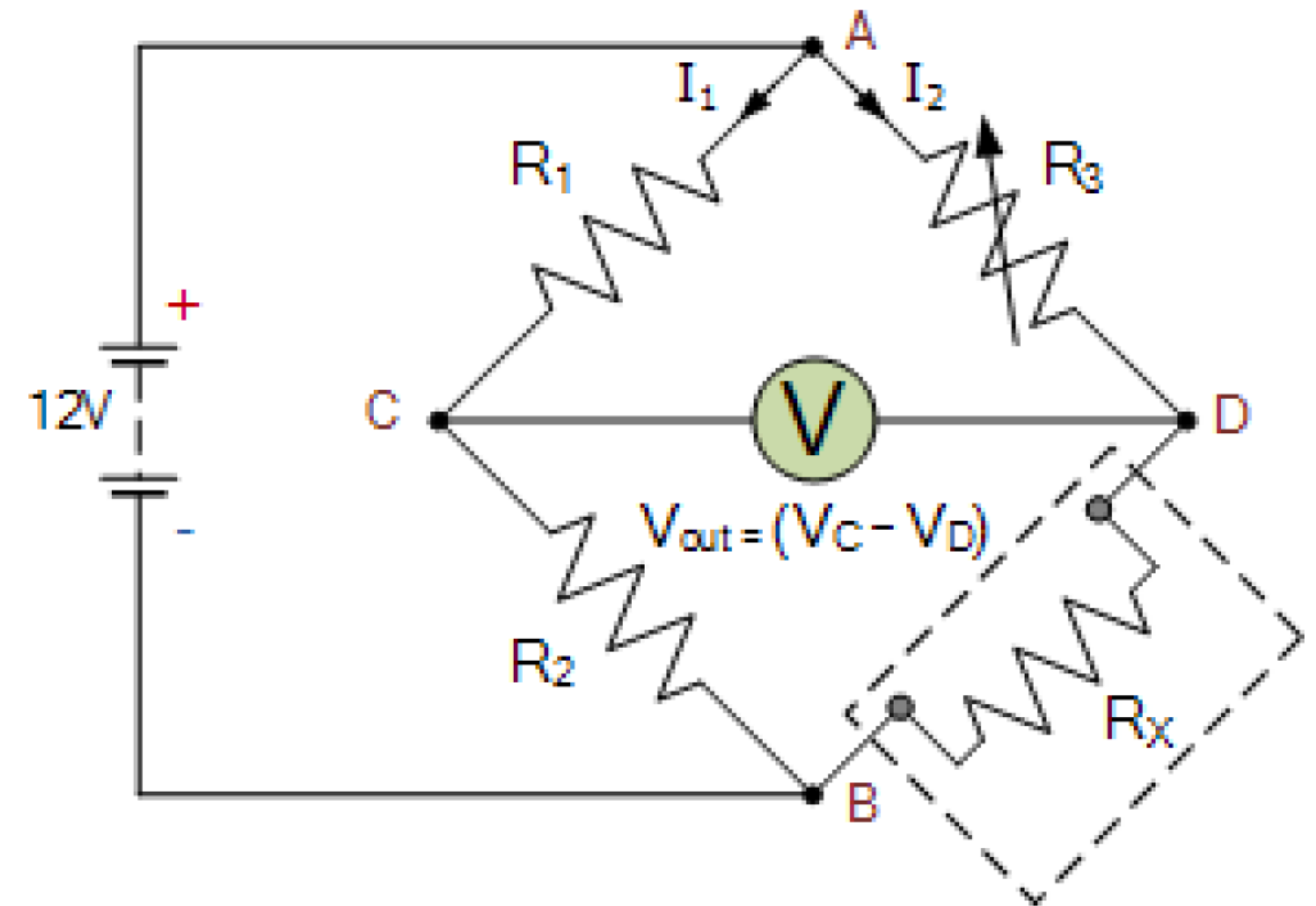
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Aplicación en reducción de resistencias.

En equilibrio, el puente cumple la regla de multiplicación de resistencias en lados opuestos:

$$R_1 \cdot R_x = R_2 \cdot R_3$$

Al no haber diferencia de potencial entre C y D, no pasará corriente por esa rama. Por lo tanto, se pueden prescindir en la reducción.



Wheatstone bridge

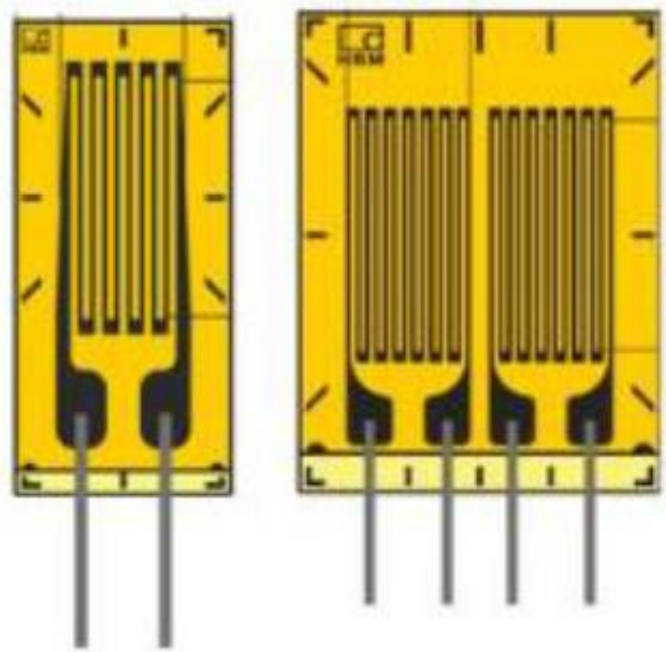
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- Medición de resistencias eléctricas desconocidas con alta precisión, especialmente en rangos bajos.
- Instrumentación electrónica: base para sensores resistivos como:
 - Galgas extensométricas (strain gauges) → detección de deformaciones mecánicas.
 - Termistores (NTC/PTC) → medición de temperatura.
 - Fotodetectores resistivos (LDR) → detección de luz.
- Compensación de variaciones externas (temperatura, envejecimiento de materiales) mediante configuraciones diferenciales.
- Circuitos de calibración y pruebas en laboratorios de electrónica y física.
- Aplicaciones industriales: monitoreo de presión, fuerza y vibraciones en sistemas de control y automatización.

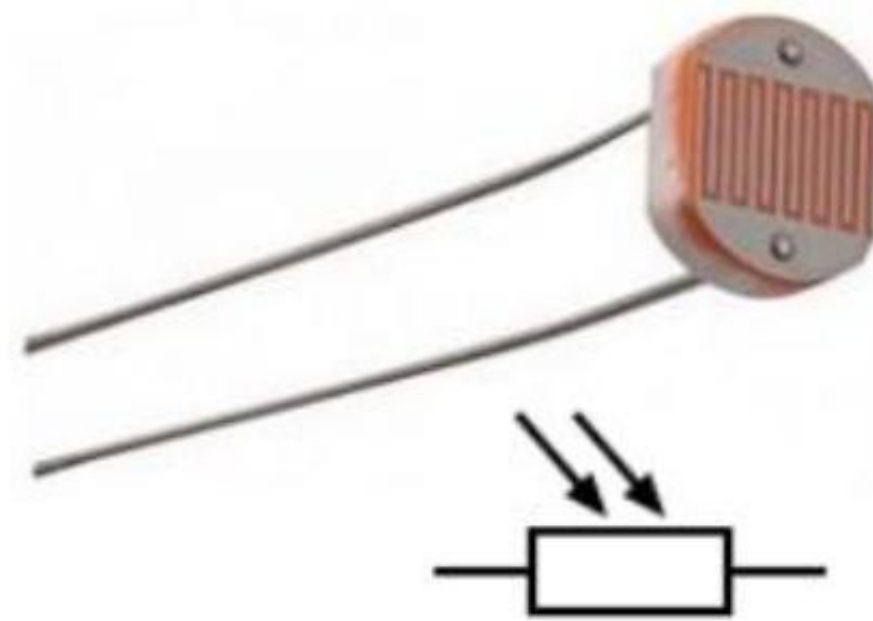
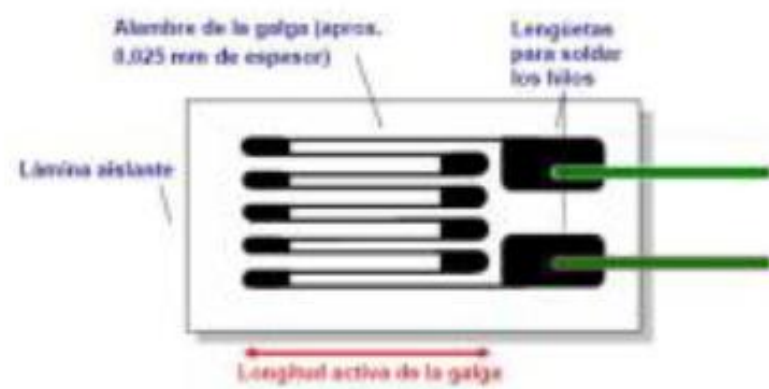
Wheatstone bridge

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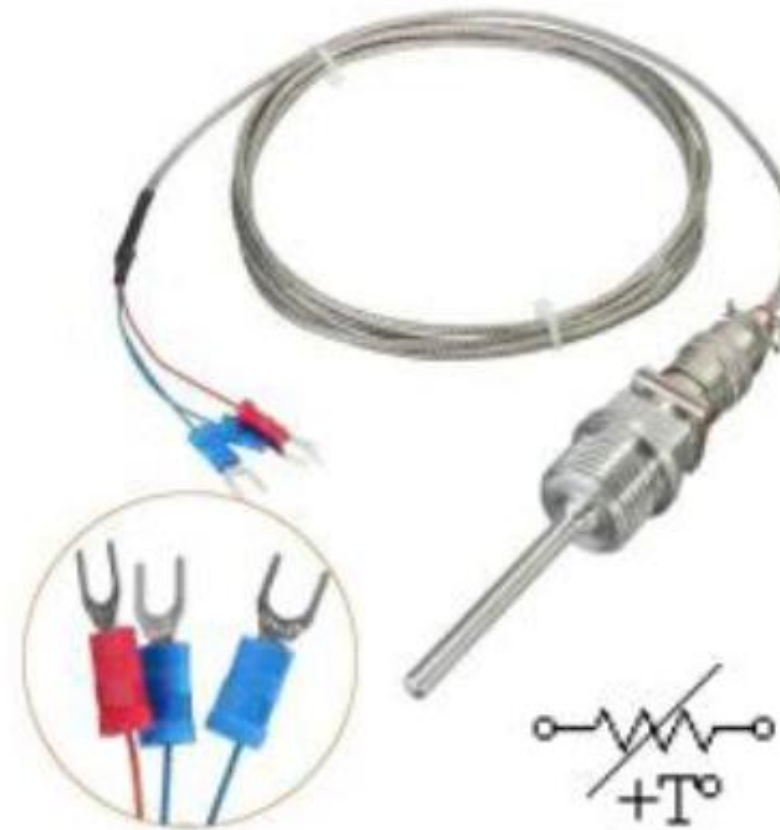
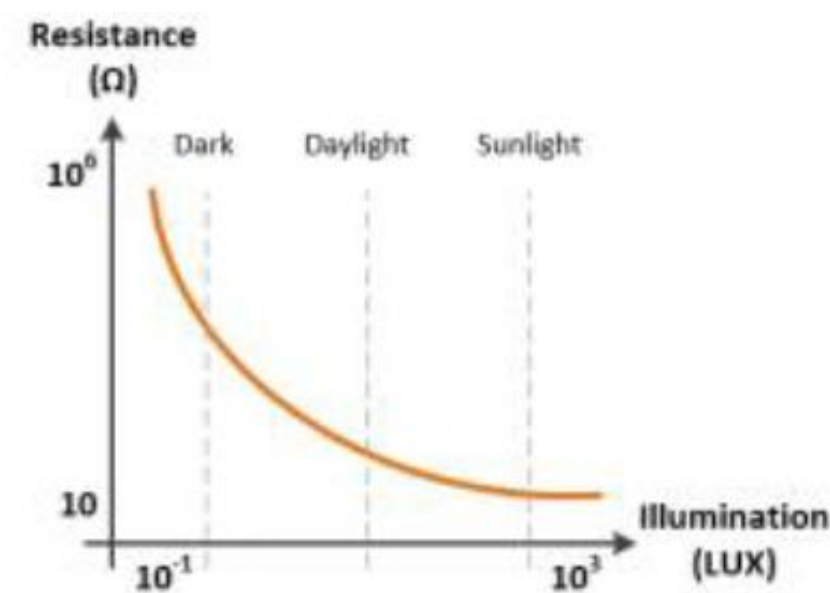
Sensores de tipo resistivo



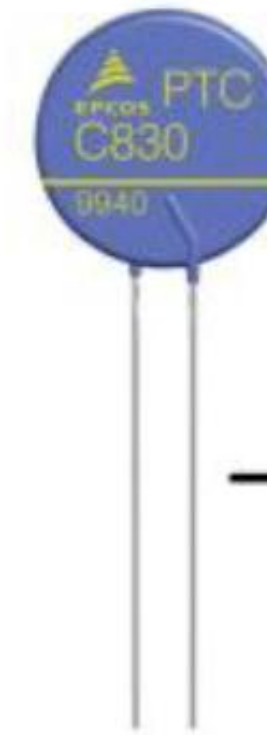
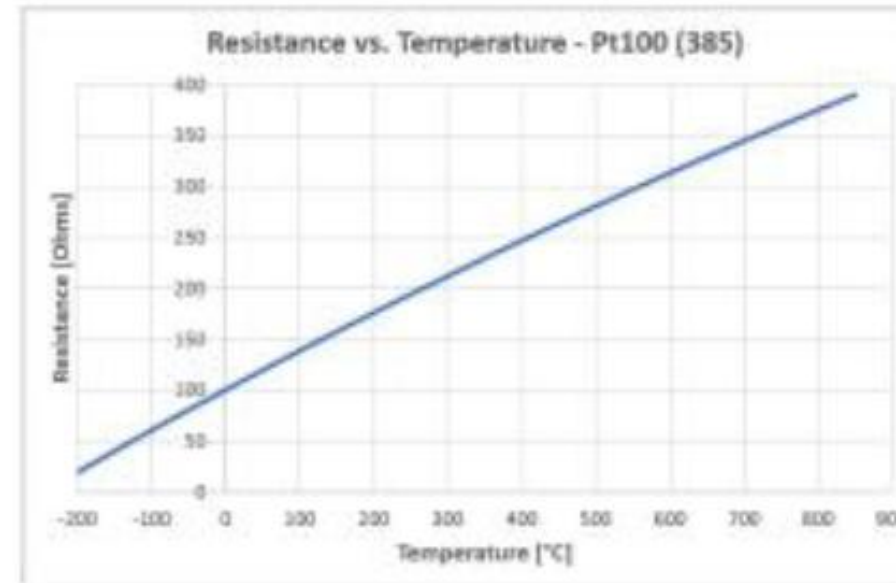
Galga extensiométrica
(strain gage)



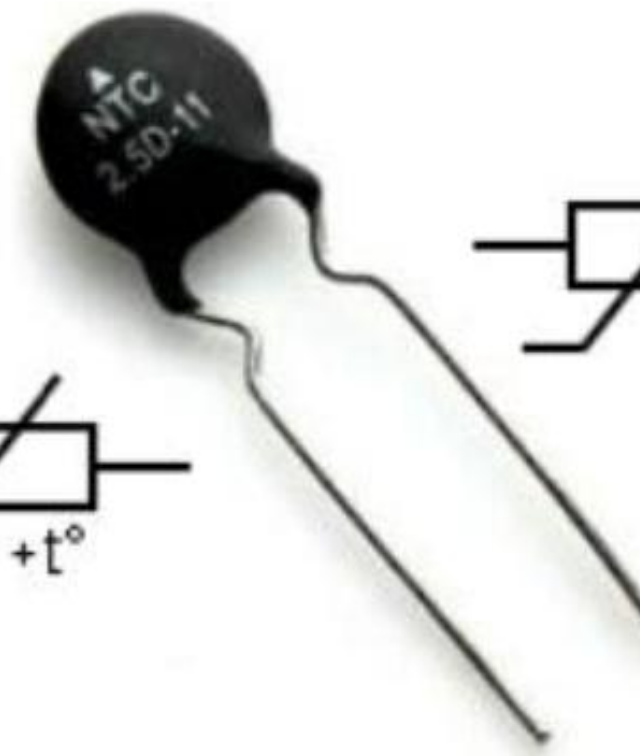
Fotorresistencia LDR
(light-dependent resistor)



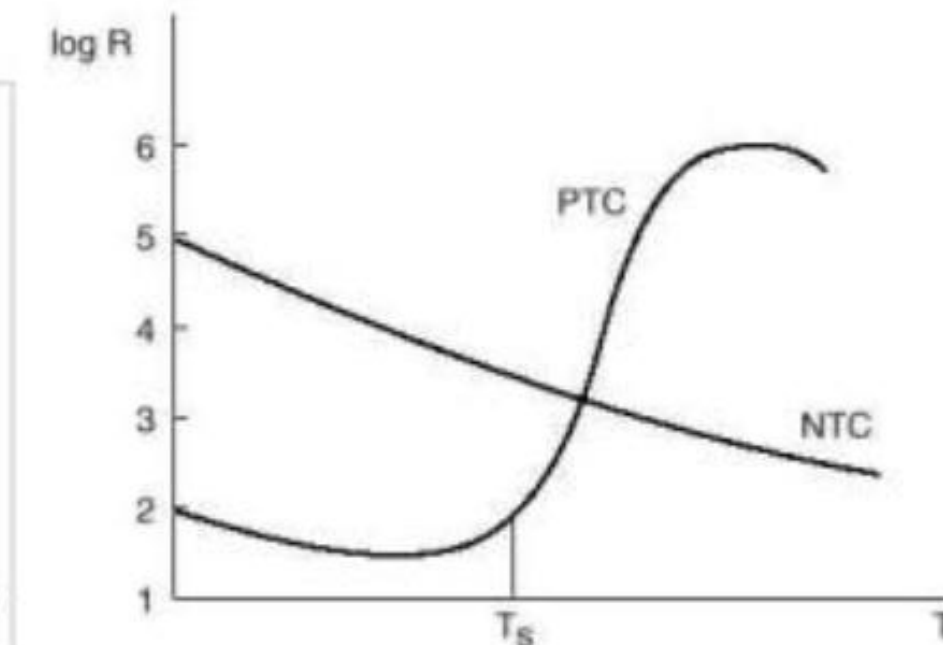
RTD PT100
todotecnologia-es.blogspot.com



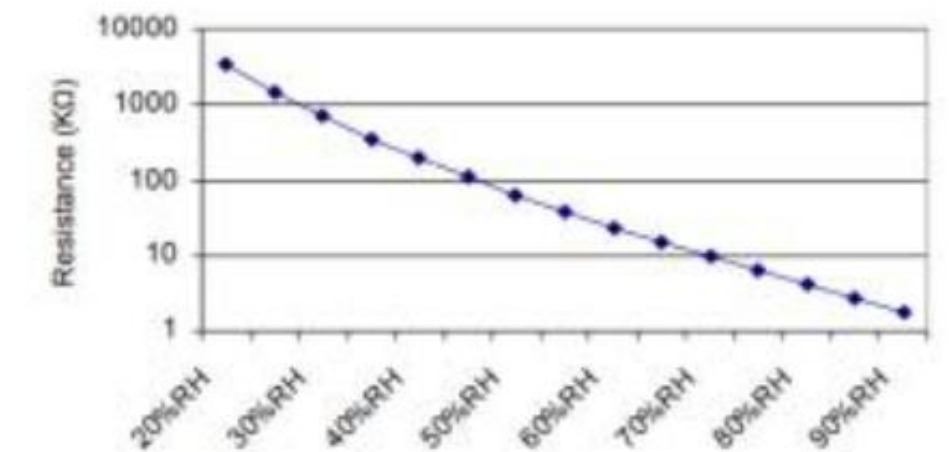
Termistor PTC



Termistor NTC



Sensor de humedad resistivo (GY-HR10x)

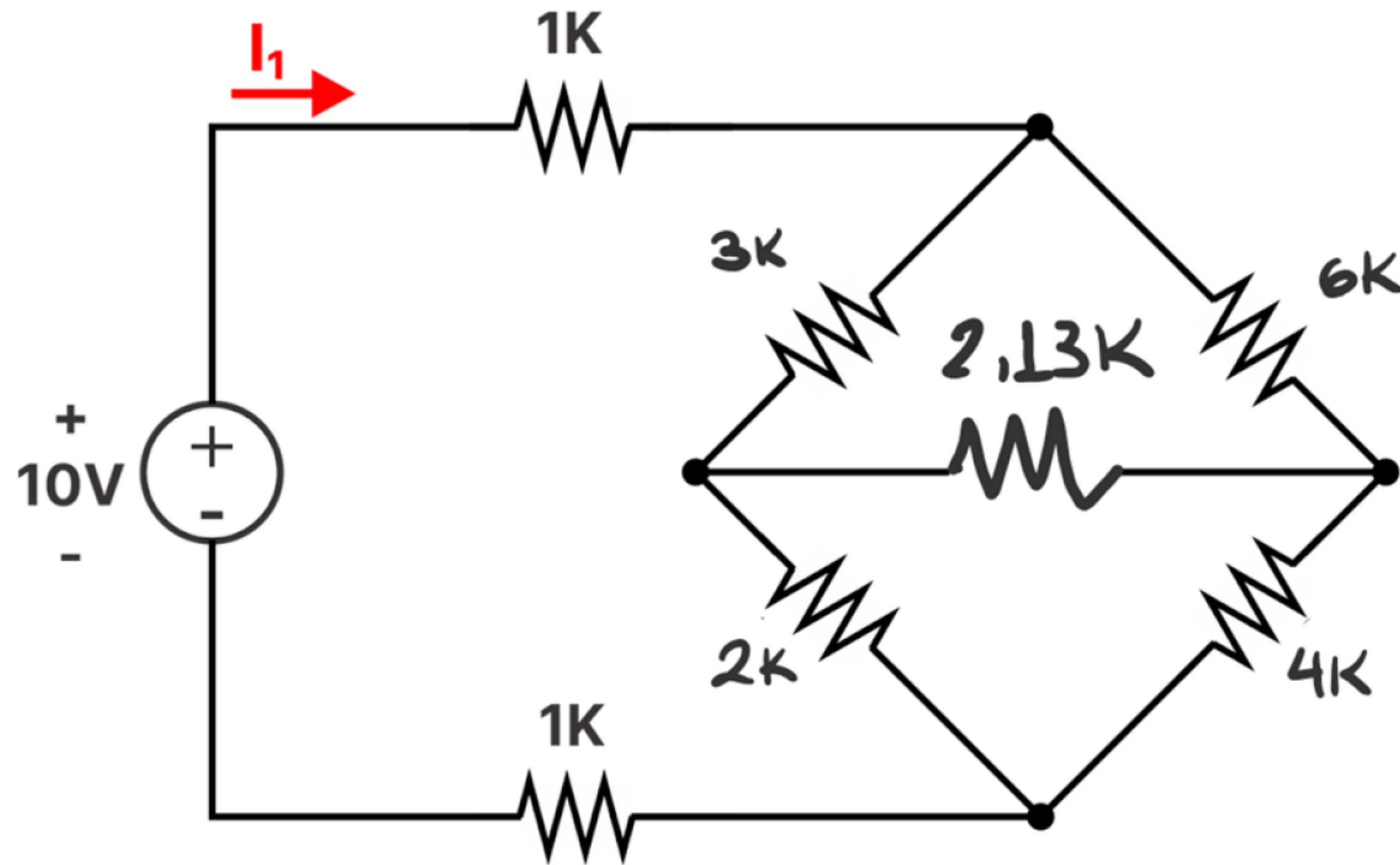


Wheatstone bridge

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5. Halle:

- (a) El valor de la corriente I_1 .
- (b) El valor de la potencia que entrega la fuente de 10V.



Outline

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Introduction

Design perspective

Sensors and actuators

Case Study

Wheatstone bridge

Conclusions

Summary

> Reinventa el mundo <

- IoT design perspectives and system approaches were discussed.
- Sensors and actuators were classified and analyzed.
- Case studies such as smart house and digital twins were reviewed.
- Wheatstone bridges were introduced in the lab.

3. Sensor Layer

Internet of Things